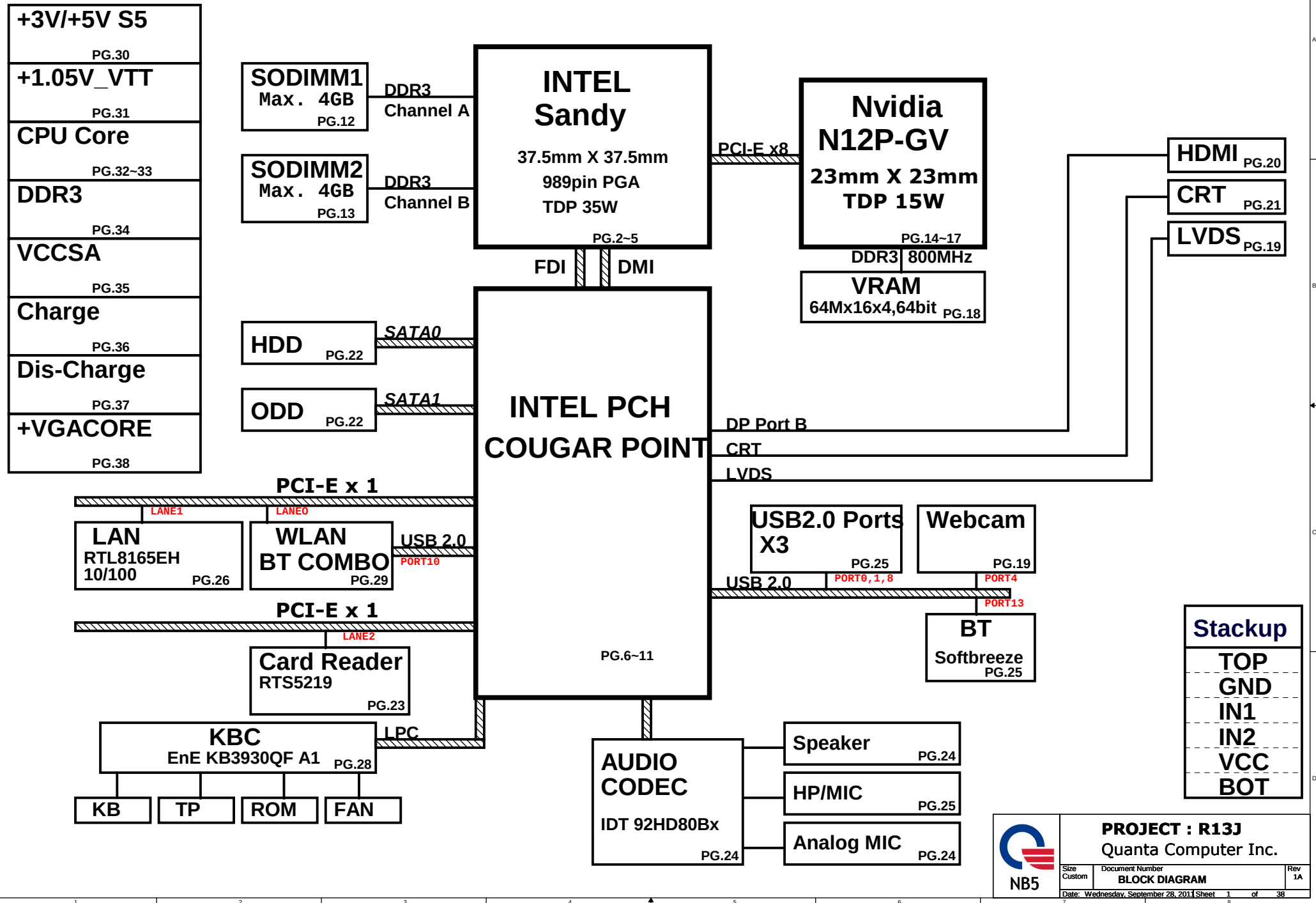


R13J INTEL UMA/DISCRETE SYSTEM DIAGRAM

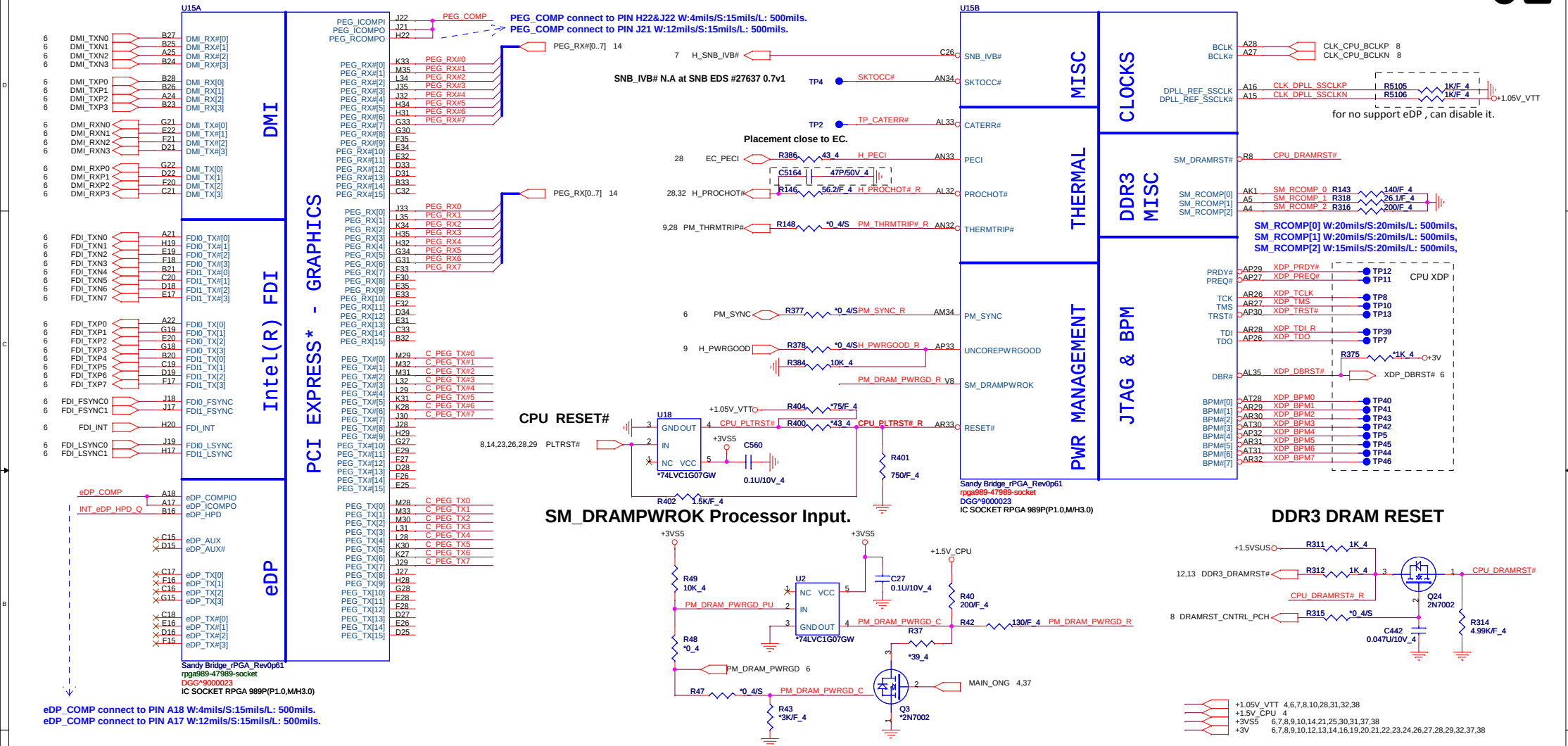
01



Sandy Bridge Processor (DMI, PEG, FDI)

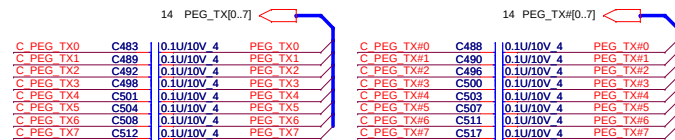
Sandy Bridge Processor (CLK, MISC, JTAG)

02

FDI disable
(DIS only stuff)

FDI_FSYNC can gang all these 4 signals together and tie them with only one 1K resistor to GND (DG V0.5 Ch2.2.9).

PEG x16 disable (UMA only remove)



0.22uF AC coupling Caps for PCIE GEN1/2/3

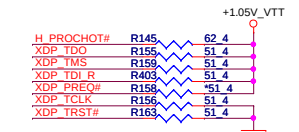
0.22uF AC coupling Caps for PCIE GEN1/2/3

DP & PEG Compensation

+1.05V_VTT R67 10K_4 INT_eDP_HPD_Q
 +1.05V_VTT R319 24.9F_4 eDP_COMP
 eDP_COMPIO and ICOMPO signals should be shorted near balls and routed with typical impedance <25 mohms
 +1.05V_VTT R85 24.9F_4 PEG_COMP

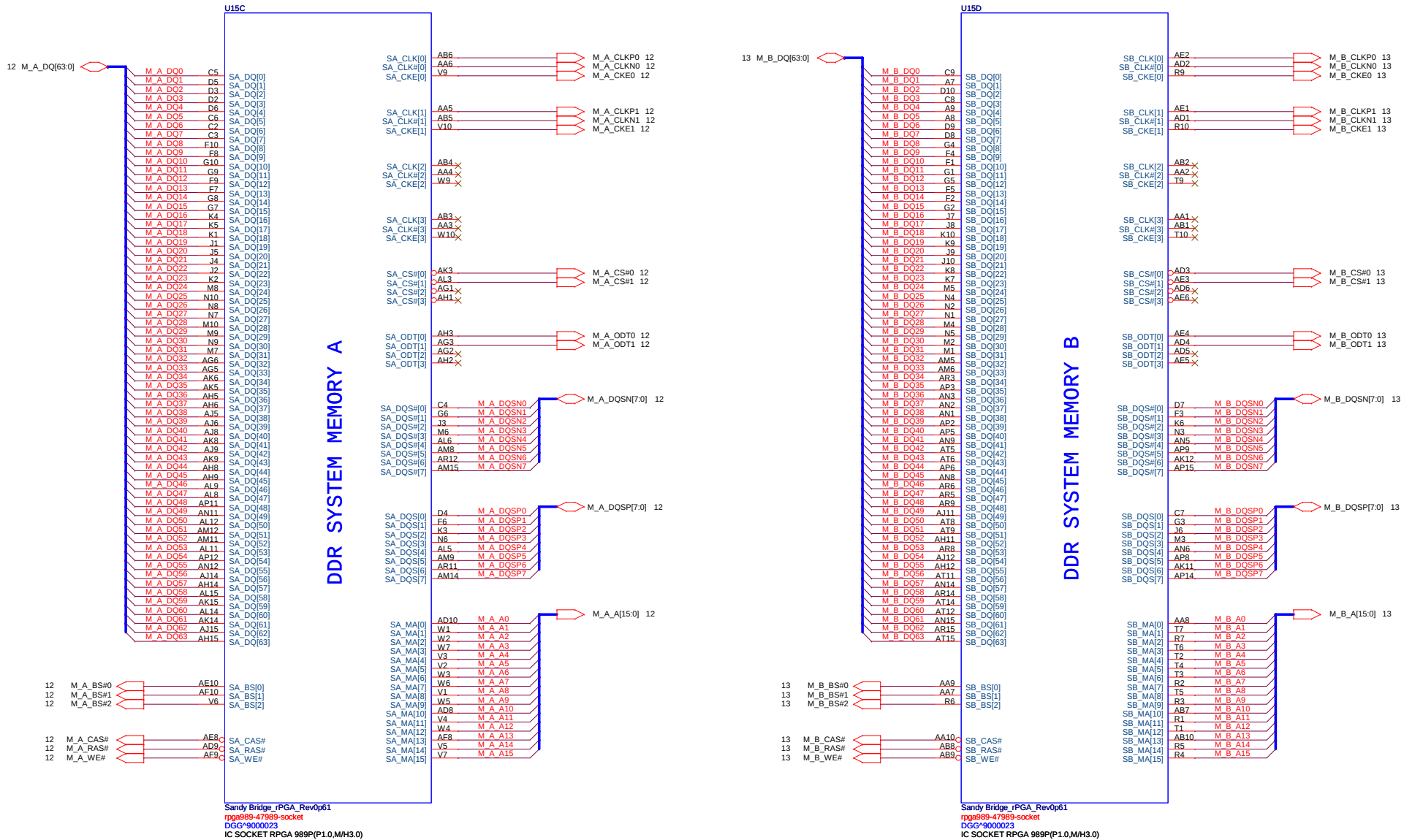
PEG_ICOMPI and RCOMP0 signals should be routed within 500 mils typical impedance = 43 mohms PEG_ICOMPO signals should be routed within 500 mils typical impedance = 14.5 mohms

Processor pull-up (CPU)

PROJECT : R13J
Quanta Computer Inc.

Size	Document Number	Rev
Custom	SNB 1/4 (PCIE&DMI&FDI)	1A
Date:	Wednesday, September 28, 2011	Sheet 2 of 38

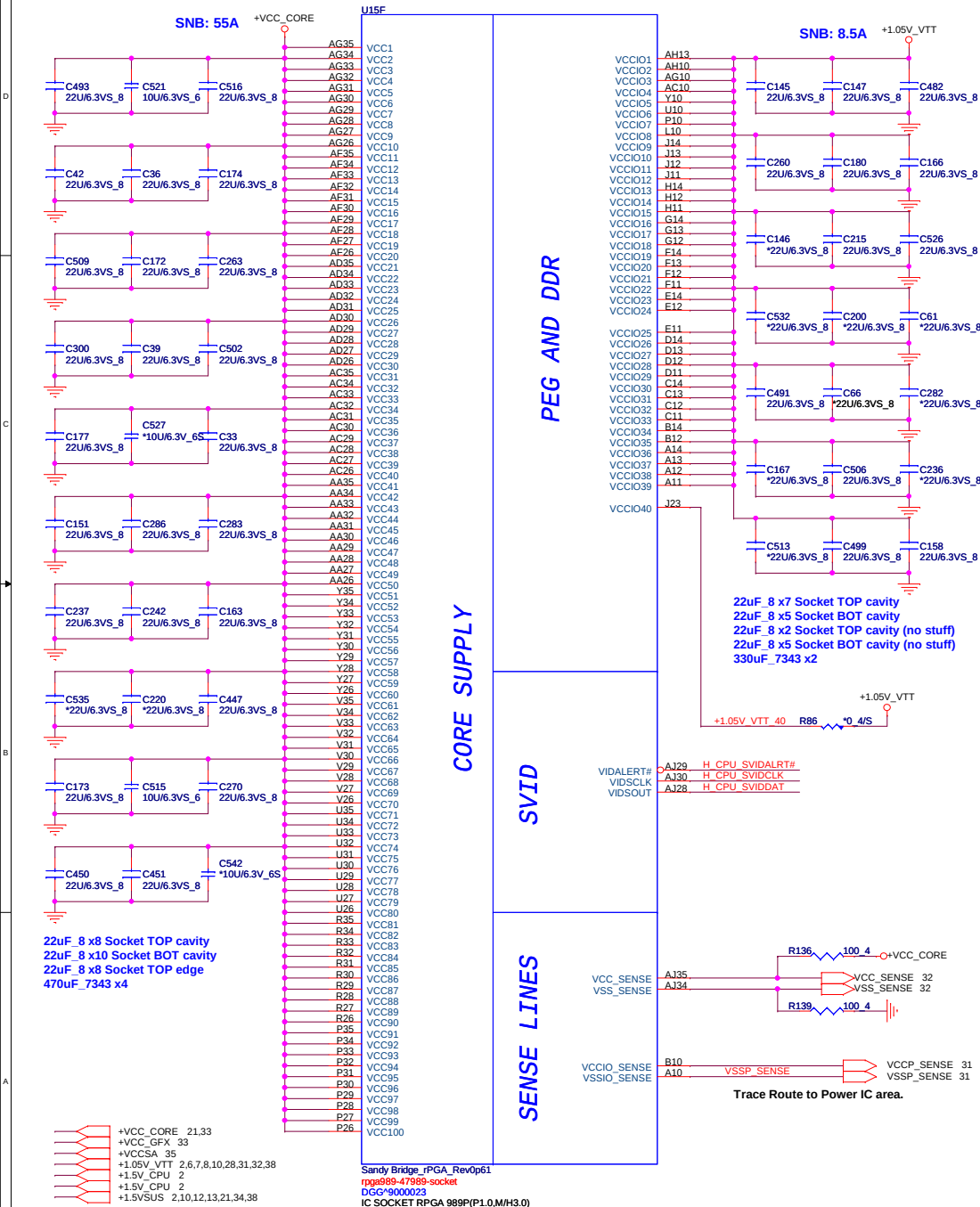
Sandy Bridge Processor (DDR3)



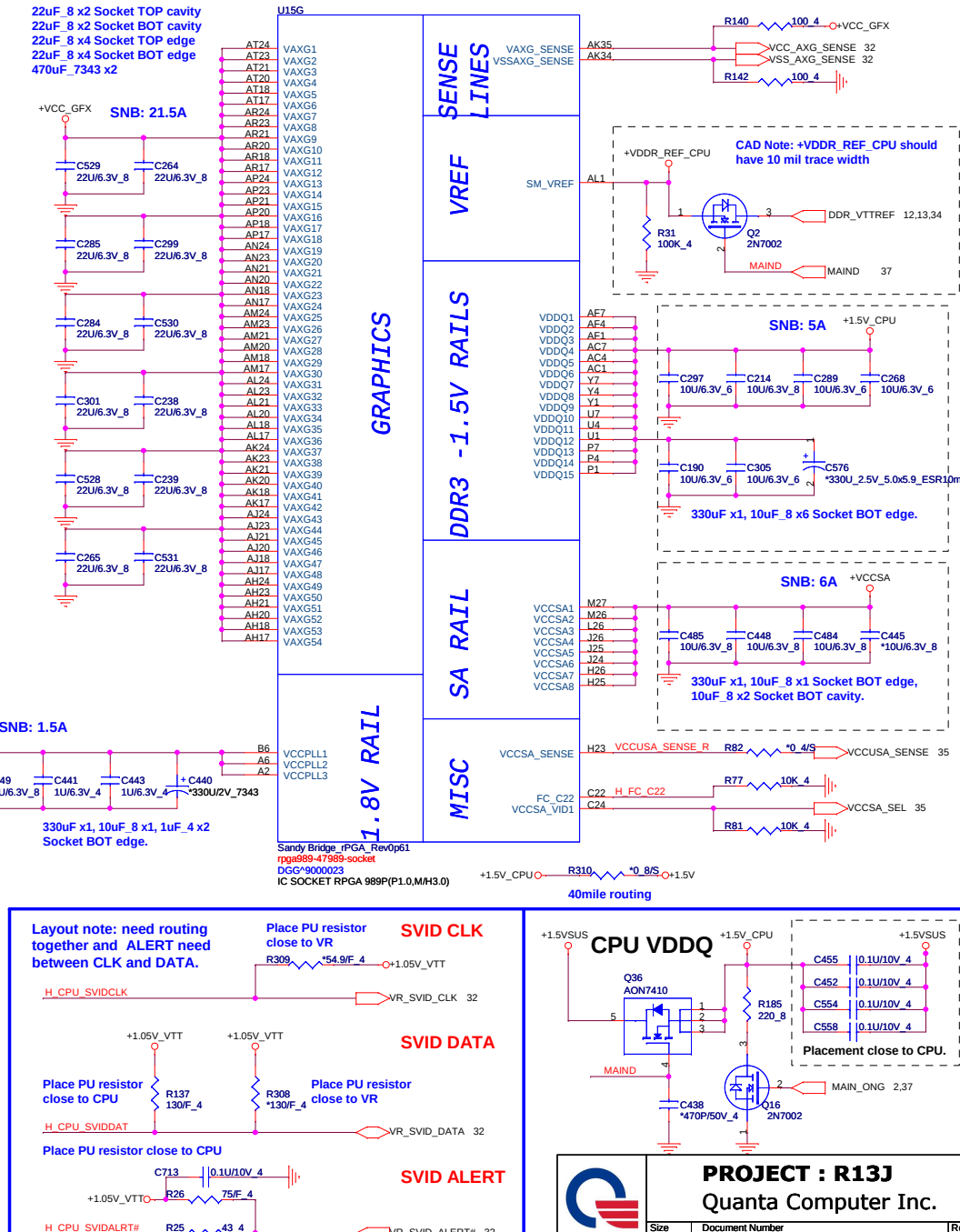
Sandy Bridge_rPGA_Rev0p61
rpga989-47989-socket
DGG^9000023
IC SOCKET RPGA 989P(P1.0,M/H3.0)

Sandy Bridge_rPGA_Rev0p61
rpga989-47989-socket
DGG^9000023
IC SOCKET RPGA 989P(P1.0,M/H3.0)

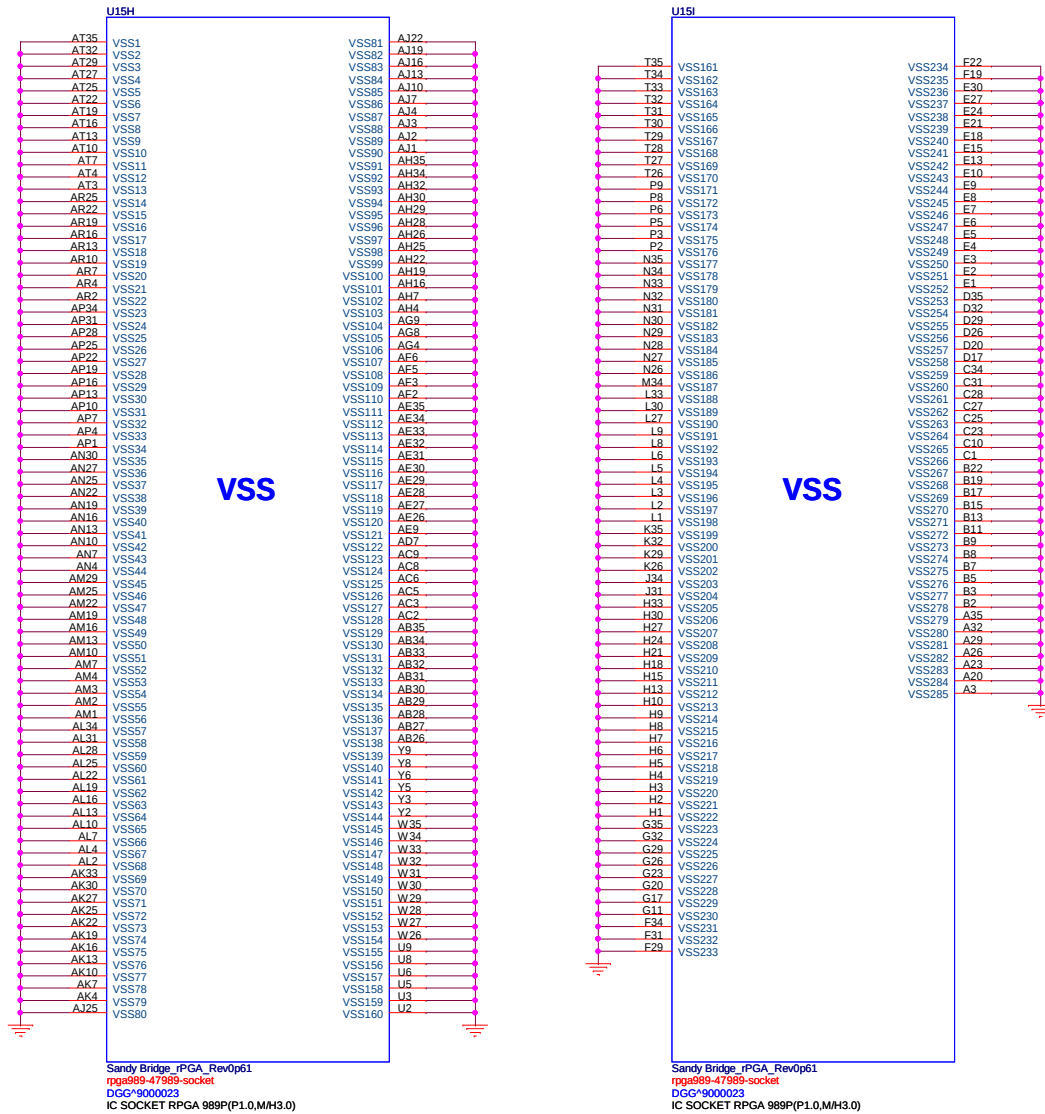
Sandy Bridge Processor (POWER)



Sandy Bridge Processor (GRAPHIC POWER)



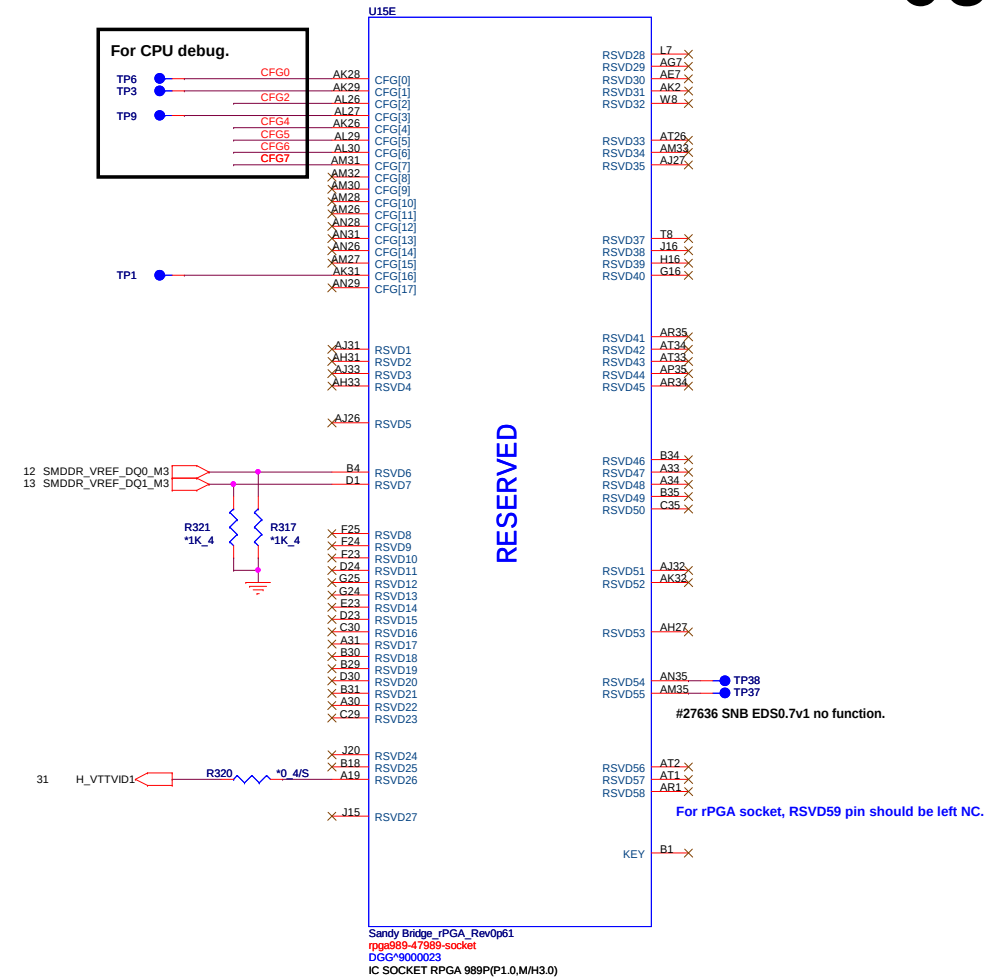
Sandy Bridge Processor (GND)



Sandy Bridge_rPGA_Rev0p61
rpg989-47989-socket
DGG*9000023
IC SOCKET RPGA 989P(P1.0,MH3.0)

Sandy Bridge_rPGA_Rev0p61
rpg989-47989-socket
DGG*9000023
IC SOCKET RPGA 989P(P1.0,MH3.0)

Sandy Bridge Processor (RESERVED, CFG)



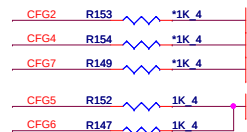
CFG[6:5] (PCIe Port Bifurcation Straps)

11: (Default) x16 - Device 1 functions 1 and 2 disabled
10: x8, x8 - Device 1 function 1 enabled ; function 2 disabled
01: Reserved - (Device 1 function 1 disabled ; function 2 enabled)
00: x8,x4,x4 - Device 1 functions 1 and 2 enabled

Processor Strapping

The CFG signals have a default value of '1' if not terminated on the board.

	1	0
CFG2 (PEG Static Lane Reversal)	Normal Operation	Lane Reversed
CFG4 (DP Presence Strap)	Disable; No physical DP attached to eDP	Enable; An ext DP device is connected to eDP
CFG7 (PEG Defer Training)	PEG train immediately following xxRESETB de assertion	PEG wait for BIOS training



PROJECT : R13J
Quanta Computer Inc.

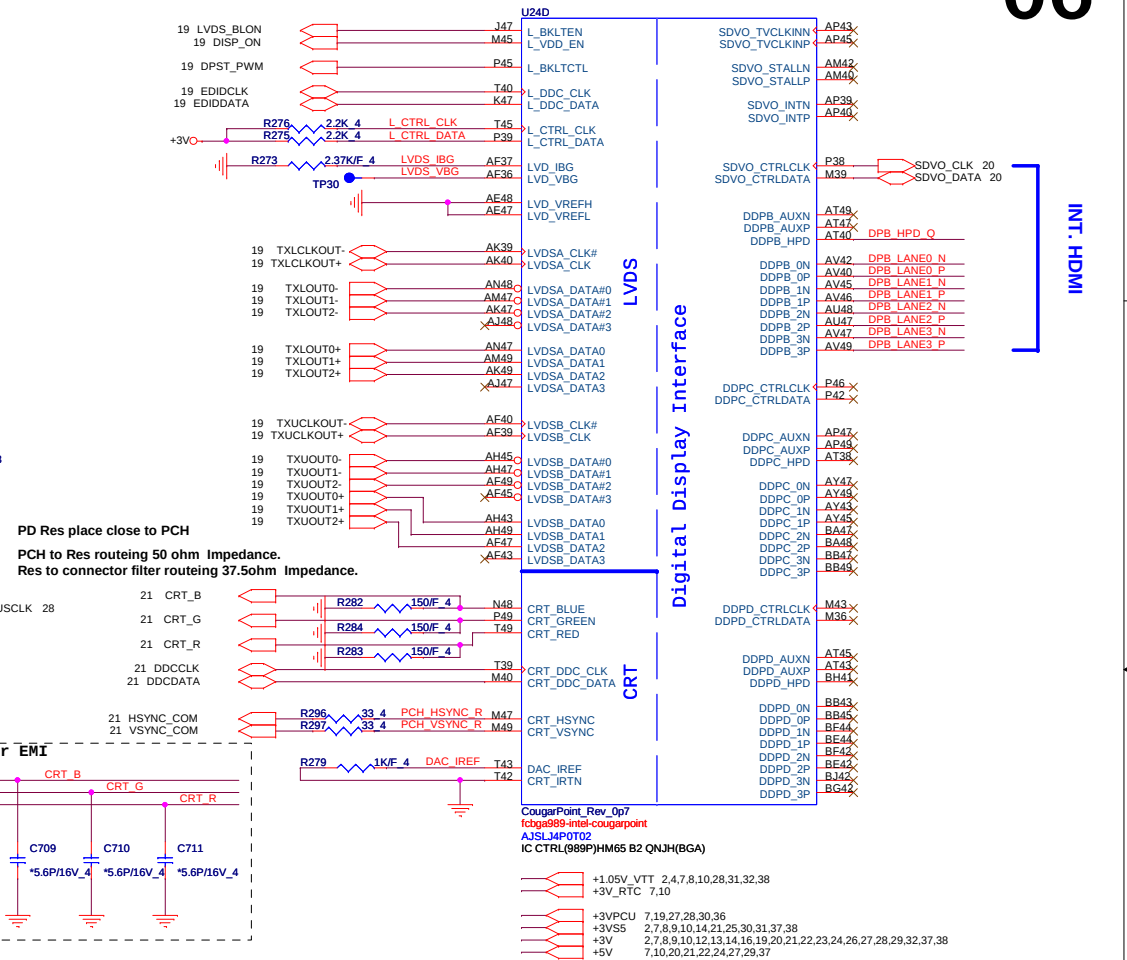
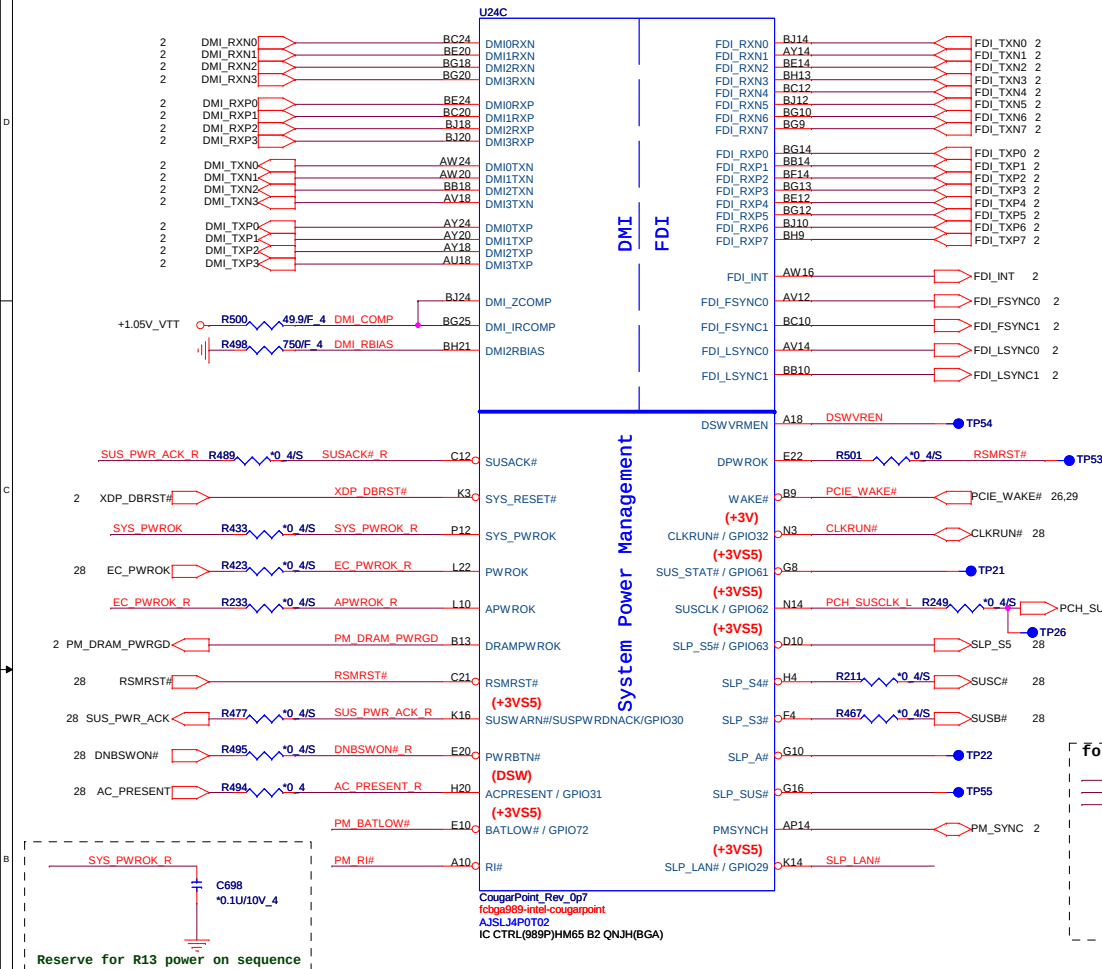
Size Custom Document Number SNB 4/4 (GND) Rev 1A

Date: Wednesday, September 28, 2011 Sheet 5 of 38

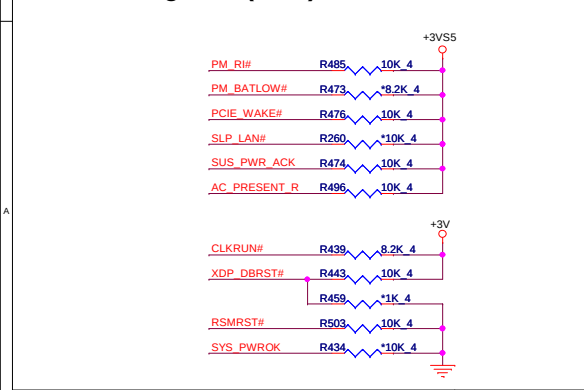
Cougar Point (DMI, FDI, PM)

Cougar Point (LVDS, DDI)

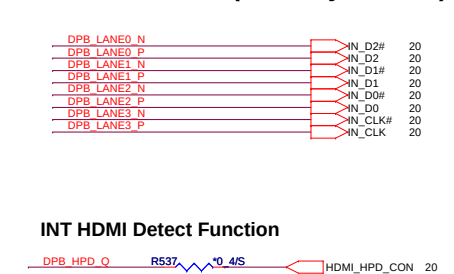
06



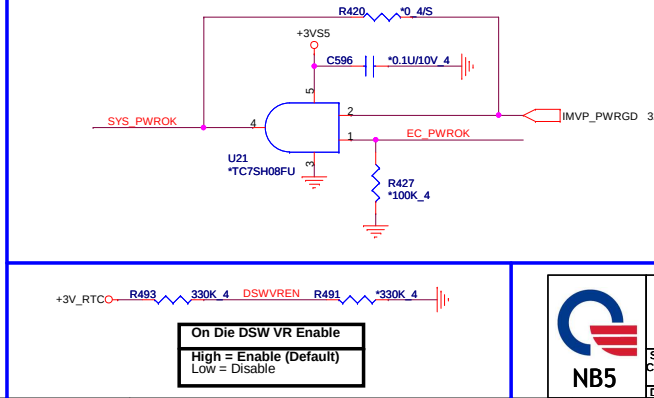
PCH Pull-high/low(CLG)



INT HDMI disable (DIS only remove)



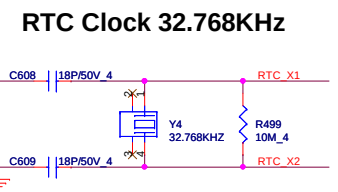
System PWR_OK(CLG)



PROJECT : R13J
Quanta Computer Inc.

Size	Document Number	Rev
Custom	PCH 1/6 (DMI/FDI/VIDEO)	1A
Date: Wednesday, September 28, 2011	Sheet 6	of 38

07



RTC Circuitry(RTC)



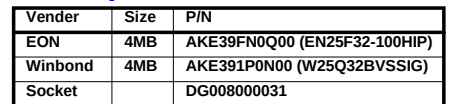
ODD (SATA1 1.5Gb/s)



PCH JTAG Debug(CLG)



PCH SPI ROM(CLG)



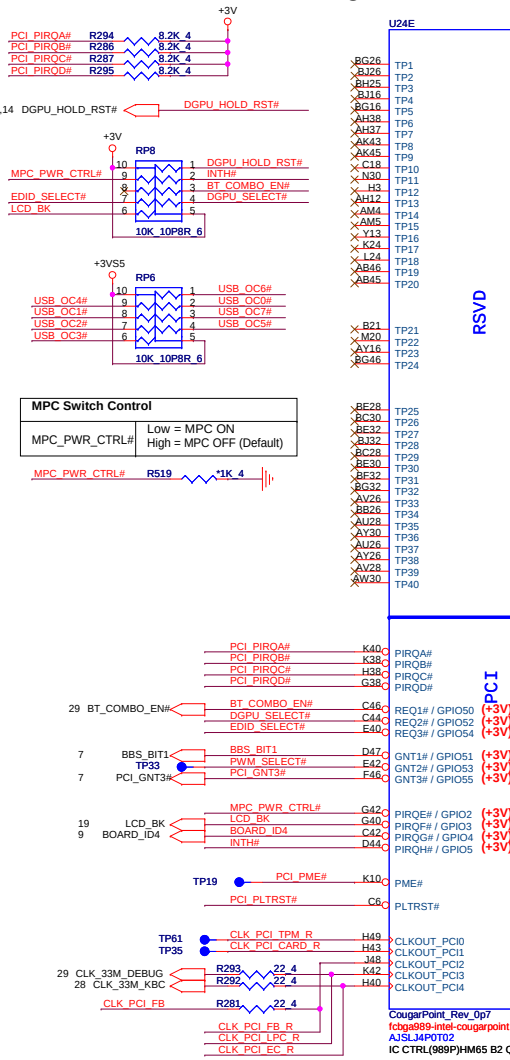
Size	Document Number	Rev
Custom	PCH 2/6 (SATA/HDA/SPI)	1A
Date: Wednesday, September 28, 2011 Sheet 7 of 38		

Pin Name	Strap description	Sampled	Configuration	Circuit									
SPKR <i>Different from Calpella</i>	No reboot mode setting	PWROK	0 = Default (weak pull-down 20K) 1 = Setting to No-Reboot mode										
GNT3# / GPIO55	Top-Block Swap Override	PWROK	0 = "top-block swap" mode 1 = Default (weak pull-up 20K)										
INTVRMEN	Integrated 1.05V VRM enable	ALWAYS	Should be always pull-up										
HDA_DOCK_EN#/GPIO33	Flash Descriptor Security <i>Only for Interposer</i>	PWROK	0 = Override 1 = Default (weak pull-up 20K)										
GNT1# / GPIO51	Boot BIOS Selection 1 [bit-1]	PWROK	<table border="1"> <thead> <tr> <th>GNT1#</th><th>GNT0#</th><th>Boot Location</th></tr> </thead> <tbody> <tr> <td>1</td><td>0</td><td>SPI</td></tr> <tr> <td>0</td><td>0</td><td>LPC</td></tr> </tbody> </table>	GNT1#	GNT0#	Boot Location	1	0	SPI	0	0	LPC	[Need external pull-down for LPC BIOS] Default weak pull-up on GNT0/1#
GNT1#	GNT0#	Boot Location											
1	0	SPI											
0	0	LPC											
GPIO19 <i>Different from Calpella</i>	Boot BIOS Selection 0 [bit-0]	PWROK											
GNT2# / GPIO53	ESI strap (Server only)	PWROK	Should not be pull-down (weak pull-up 20K)	USE GPIO PIN									
NV_ALE	Intel Anti-Theft HDD protection <i>Only for Interposer</i>	PWROK	0 = Disable (Internal pull-down 20kohm)										
NV_CLE	DMI Termination voltage	PWROK	weak pull-down 20kohm										
HDA_SYNC	On-Die PLL VR Voltage Select	RSMRST	0 = Support by 1.8V (weak pull-down) 1 = Support by 1.5V										
HDA_SDO	Flash Descriptor Security	PWROK	0 = Override 1 = Default (weak pull-up 20K)										
GPIO8	Integrated Clock Chip Enable	RSMRST#	Should be pull-down (weak pull-up 20K)										
GPIO28 <i>Different from Calpella</i>	On-die PLL Voltage Regulator	RSMRST#	0 = Disable 1 = Enable (Default)										
SPI_MOSI	iTPM function Disable	APWROK	0 = Default (weak pull-down 20K) 1 = Enable										

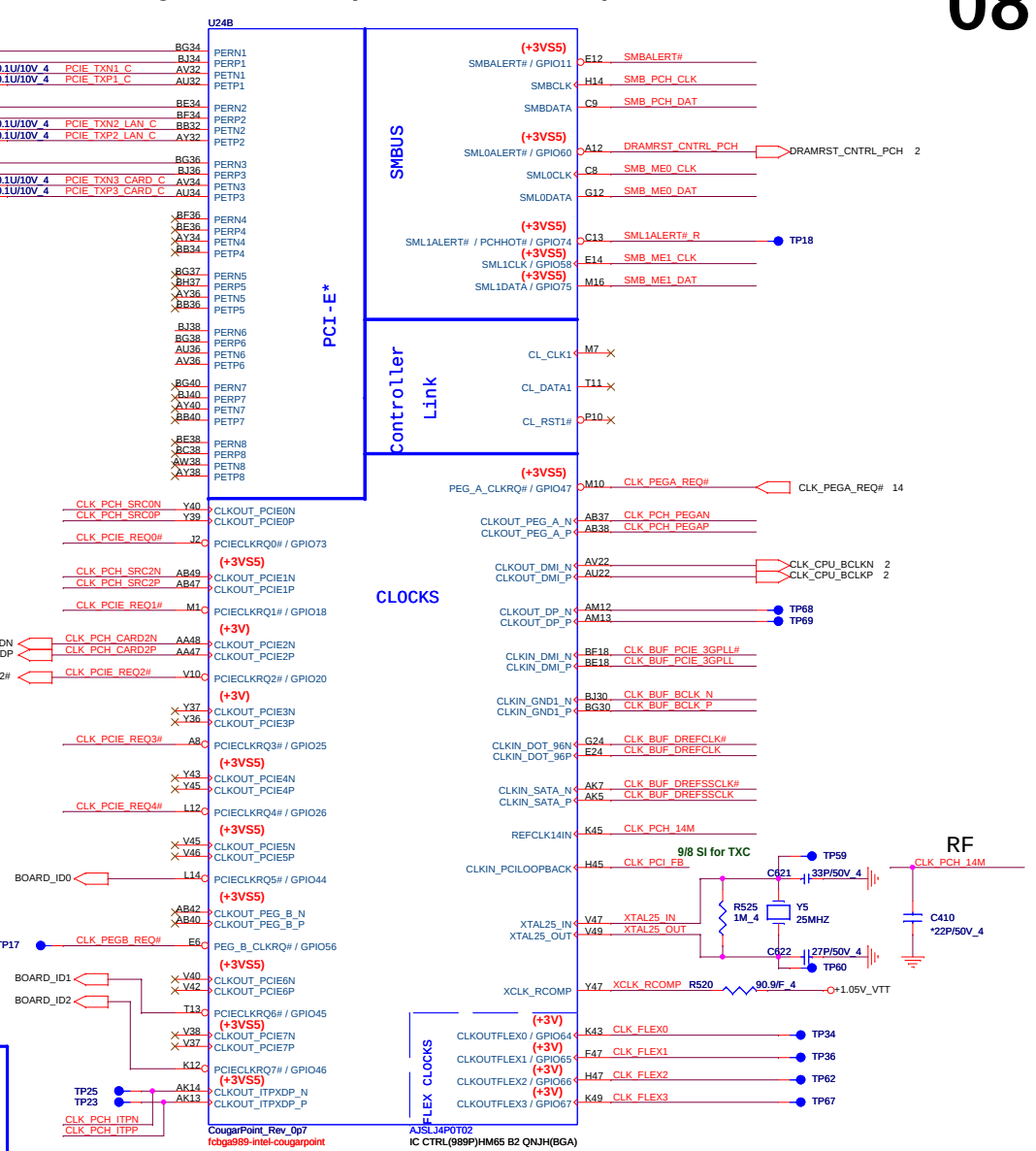
Pin Name	Strap description	Sampled	Configuration	Circuit									
SPKR Different from Calpella	No reboot mode setting	PWROK	0 = Default (weak pull-down 20K) 1 = Setting to No-Reboot mode										
GNT3# / GPIO55	Top-Block Swap Override	PWROK	0 = "top-block swap" mode 1 = Default (weak pull-up 20K)										
INTVRMEN	Integrated 1.05V VRM enable	ALWAYS	Should be always pull-up										
HDA_DOCK_EN#/GPIO33	Flash Descriptor Security Only for Interposer	PWROK	0 = Override 1 = Default (weak pull-up 20K)										
GNT1# / GPIO51	Boot BIOS Selection 1 [bit-1]	PWROK	<table><thead><tr><th>GNT1#</th><th>GNT0#</th><th>Boot Location</th></tr></thead><tbody><tr><td>1</td><td>0</td><td>SPI</td></tr><tr><td>0</td><td>0</td><td>LPC</td></tr></tbody></table>	GNT1#	GNT0#	Boot Location	1	0	SPI	0	0	LPC	[Need external pull-down for LPC BIOS] Default weak pull-up on GNT0/1#
GNT1#	GNT0#	Boot Location											
1	0	SPI											
0	0	LPC											
GPIO19 Different from Calpella	Boot BIOS Selection 0 [bit-0]	PWROK											
GNT2# / GPIO53	ESI strap (Server only)	PWROK	Should not be pull-down (weak pull-up 20K)	USE GPIO PIN									
NV_ALE	Intel Anti-Theft HDD protection Only for Interposer	PWROK	0 = Disable (Internal pull-down 20kohm)										
NV_CLE	DMI Termination voltage	PWROK	weak pull-down 20kohm										
HDA_SYNC	On-Die PLL VR Voltage Select	RSMRST	0 = Support by 1.8V (weak pull-down) 1 = Support by 1.5V										
HDA_SDO	Flash Descriptor Security	PWROK	0 = Override 1 = Default (weak pull-up 20K)										
GPIO8	Integrated Clock Chip Enable	RSMRST#	Should be pull-down (weak pull-up 20K)										
GPIO28 Different from Calpella	On-die PLL Voltage Regulator	RSMRST#	0 = Disable 1 = Enable (Default)										
SPI_MOSI	iTPM function Disable	APWROK	0 = Default (weak pull-down 20K) 1 = Enable										

PCI/USB0C# Pull-up(CLG)

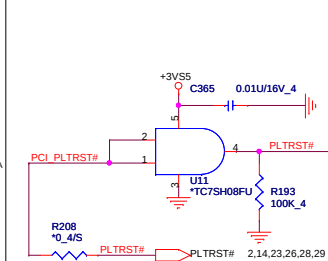
Cougar Point-M (PCI, USB, NVRAM)



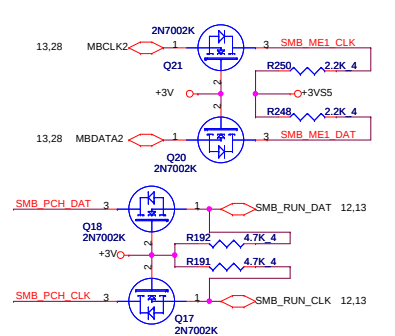
Cougar Point-M (PCI-E, SMBUS, CLK)



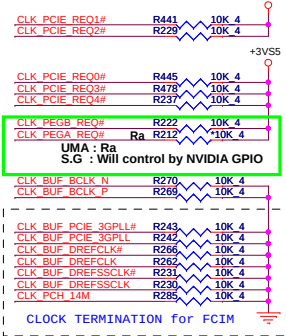
PLTRST#(CLG)



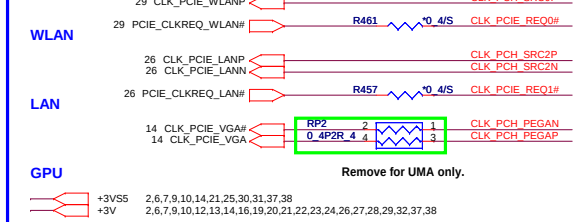
SMBus/Pull-up(CLG)



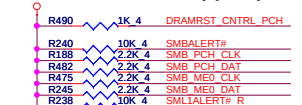
CLK_REQ/Strap Pin(CLG)



PCIE Clock

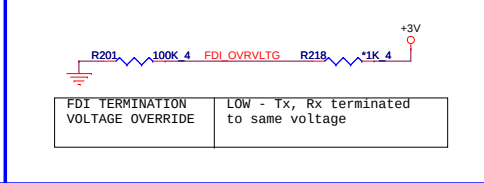
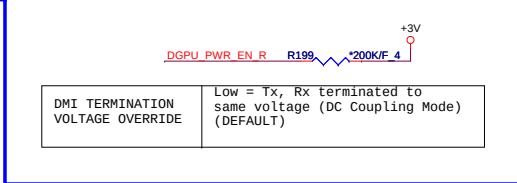
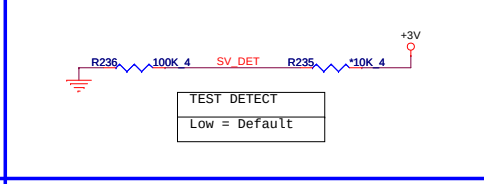
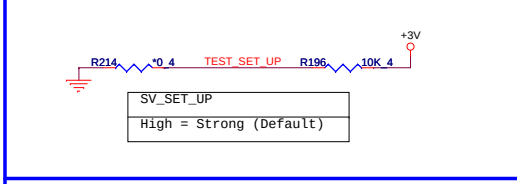
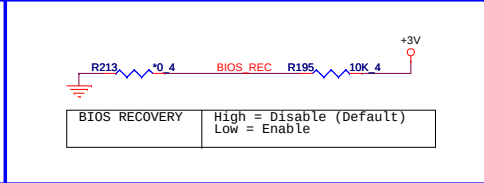
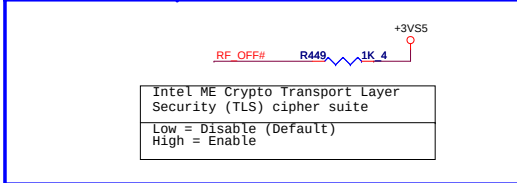
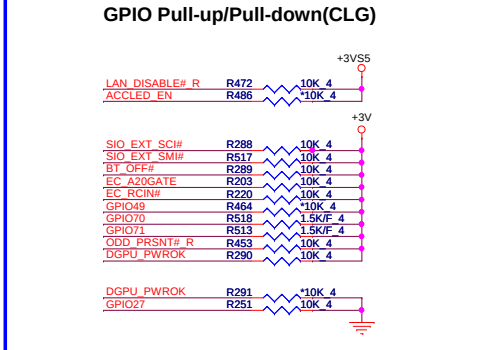
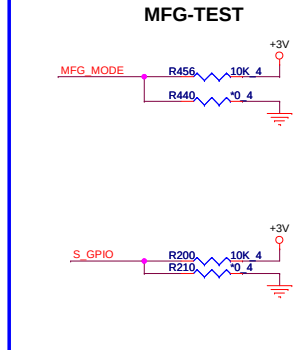
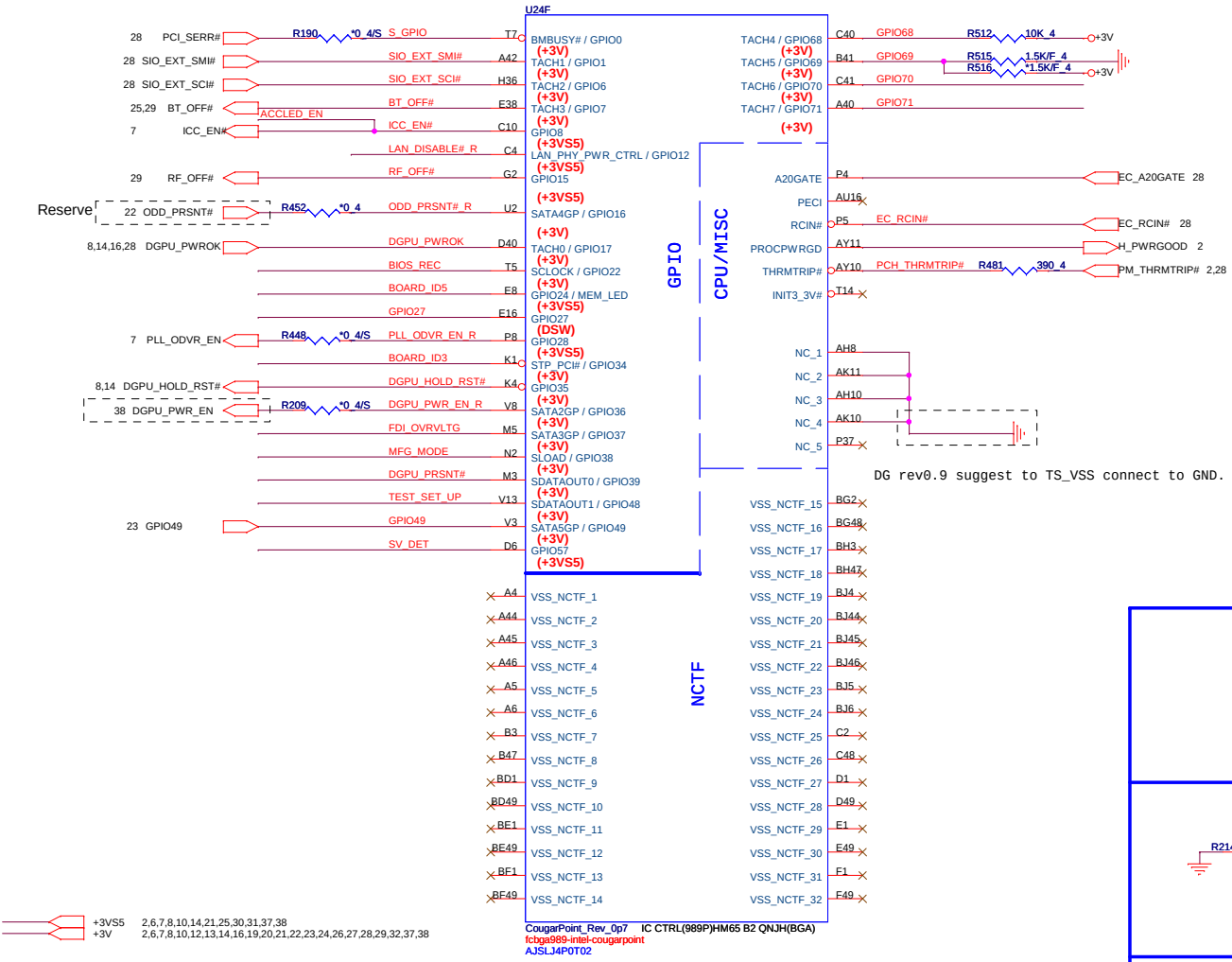


SMBus/Pull-up(CLG)



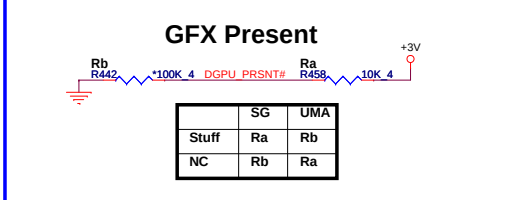
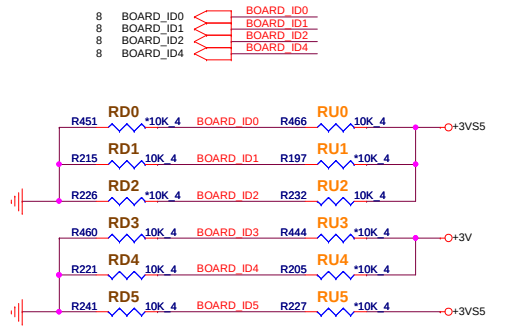
Cougar Point (GPIO,VSS_NCTF,RSVD)

Clock Gen Power OK (CLG)



BOARD ID SETTING

Model	BOARD_ID5	BOARD_ID4	BOARD_ID3	BOARD_ID2	BOARD_ID1	BOARD_ID0
R13 UMA	0	0	0	0	0	0
R13 DIS	0	0	0	0	0	1
R13 (I.1) DIS (AMD)	0	0	0	0	1	1
R13J (I.3) DIS (NVIDIA)	0	0	0	1	0	1
	0	0	0	0	0	0



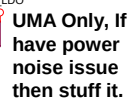
PROJECT : R13J
Quanta Computer Inc.

Size	Document Number	Rev
Custom	PCH 4/6 (GPIO/MISC)	1A
Date: Wednesday, September 28, 2011 Sheet 9 of 38		

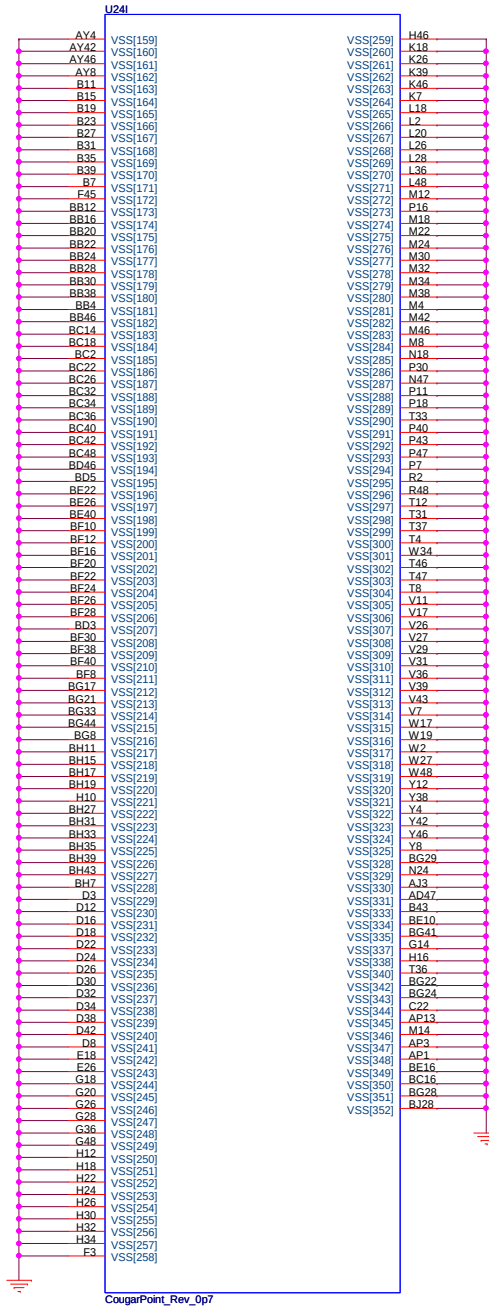
3V 10



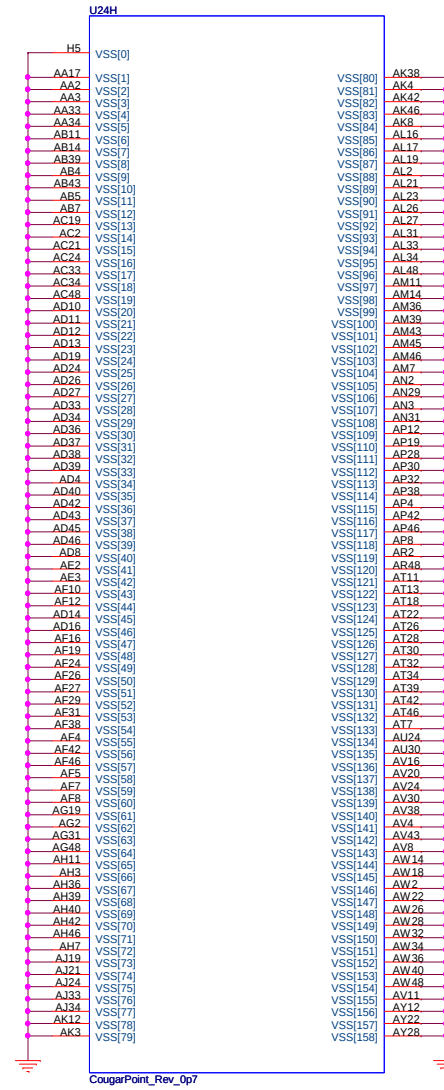
3V 10



IBEX PEAK-M (GND)



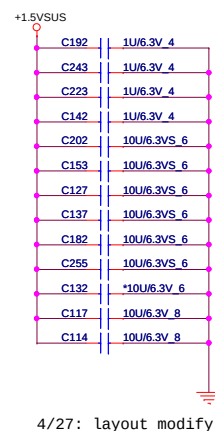
IBEX PEAK-M (GND)





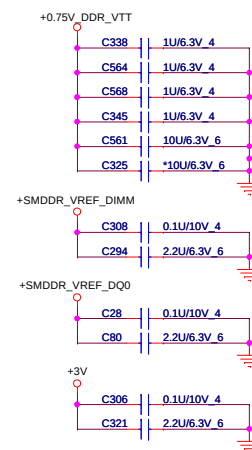
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del M2 solution
```

VREF DQ0 M2 Solution

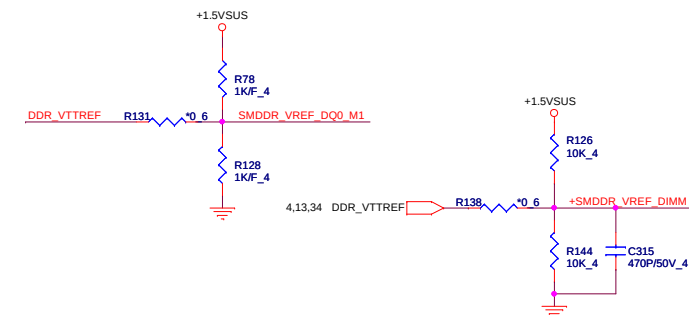


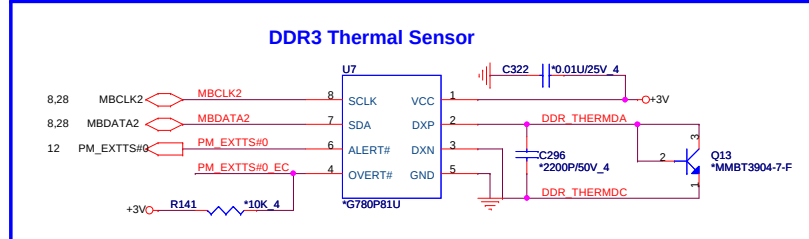
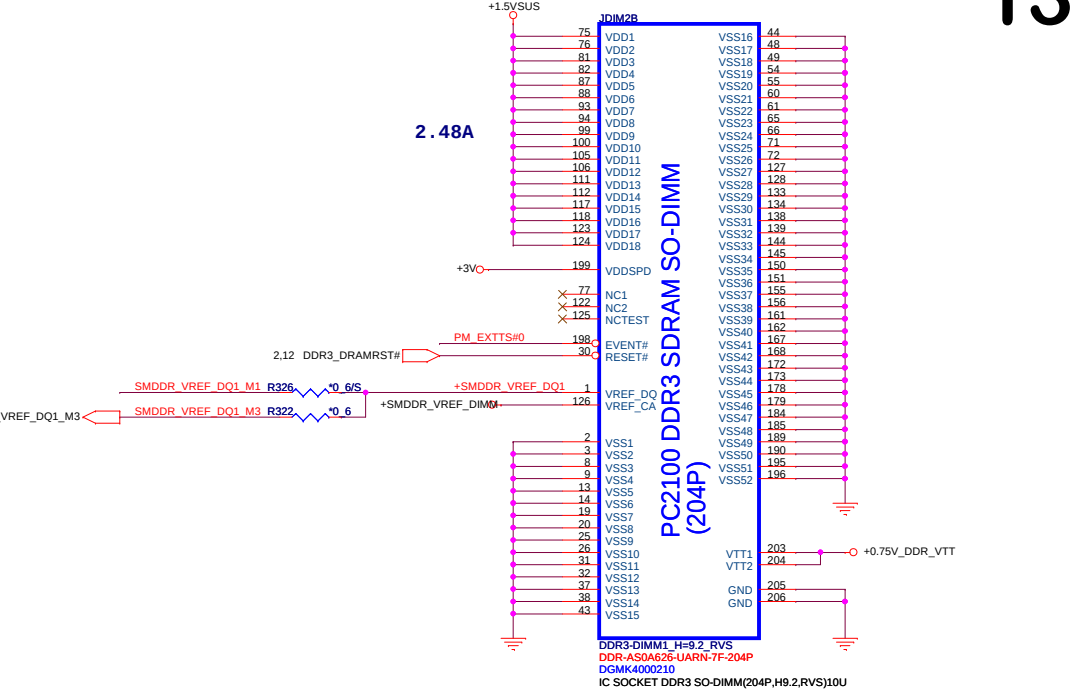
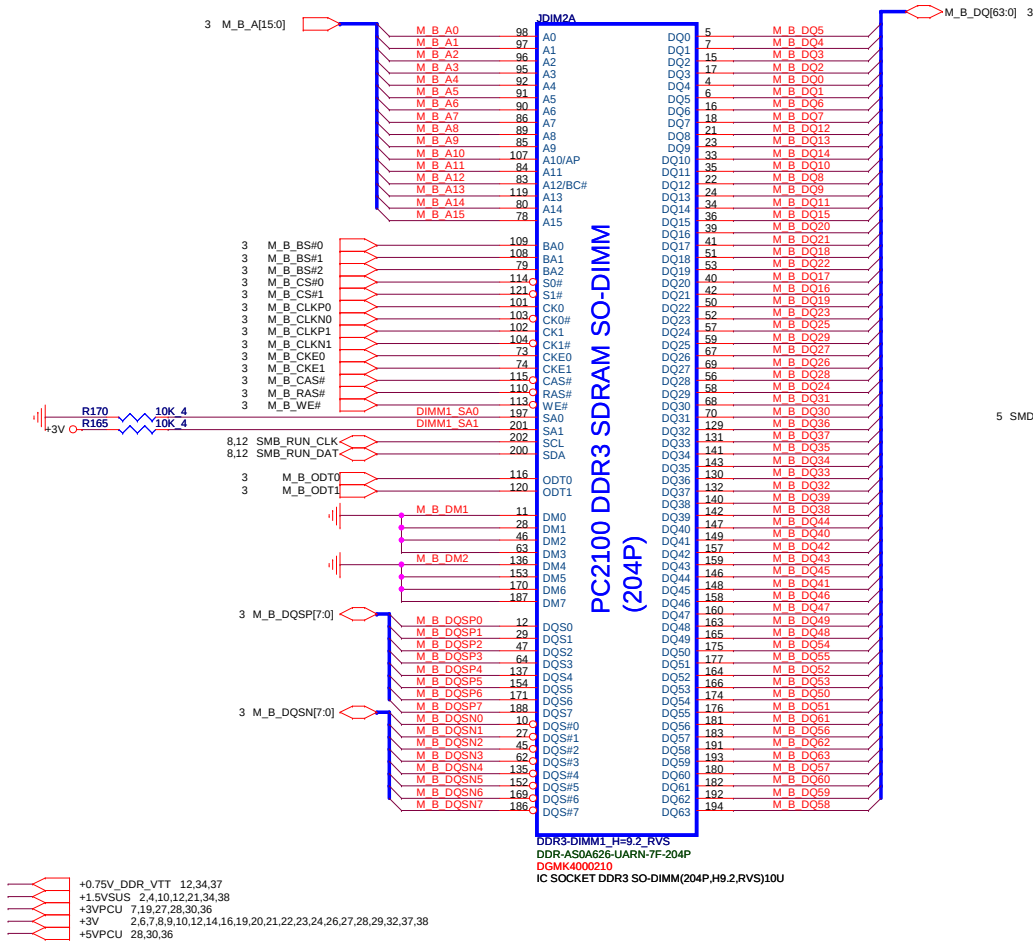
4/27: layout modify

Place these Caps near So-Dimm0.



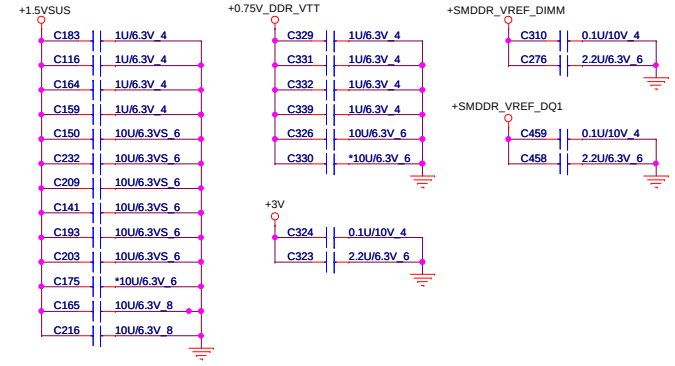
VREF DQ0 M1 Solution



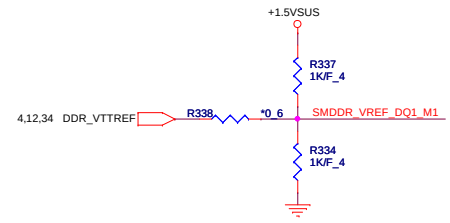


del M2 solution

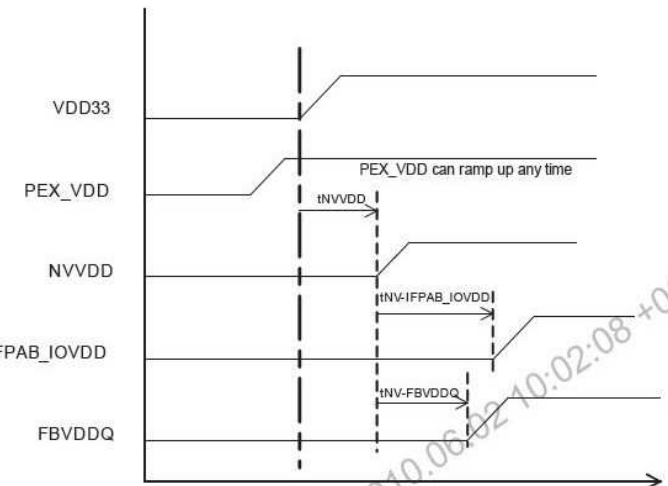
Place these Caps near So-Dimm1.



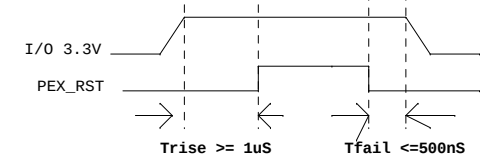
VREF DQ1 M1 Solution



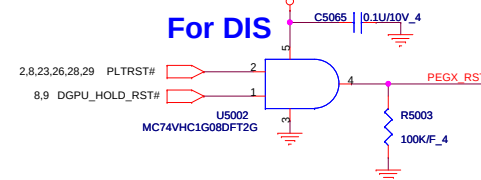
power up sequence



PEX_RST timing

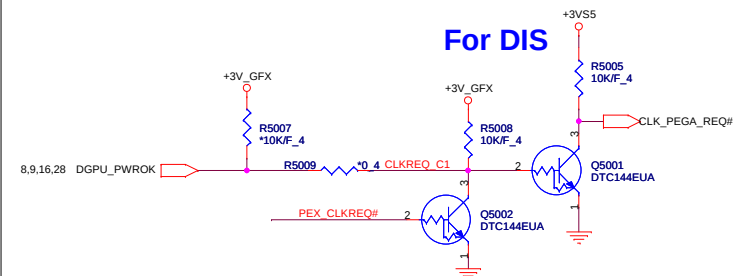


For DIS



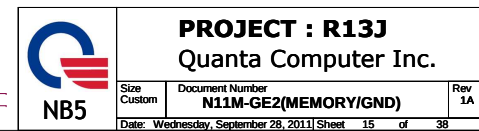
12/28 Nvidia to suggest R5009 not stuff and R5008 and Q5002 stuff.

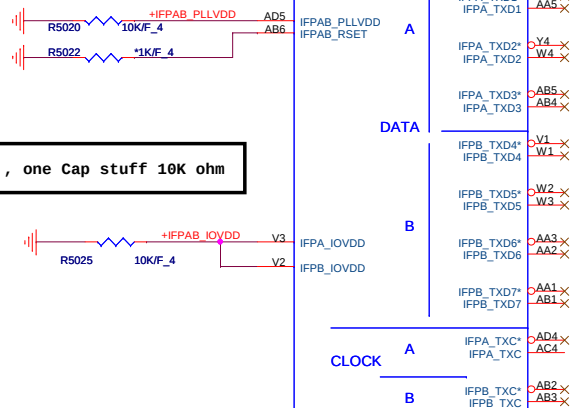
For DIS



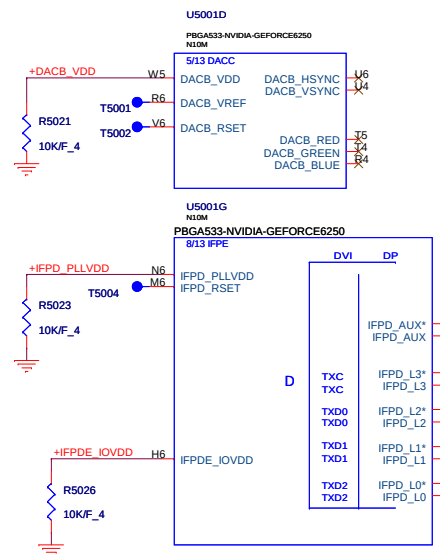
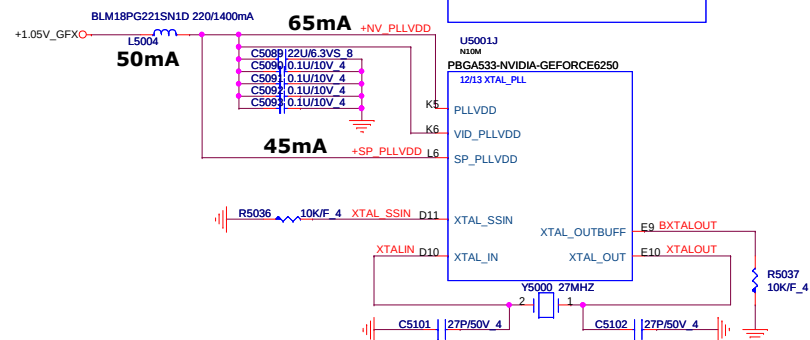
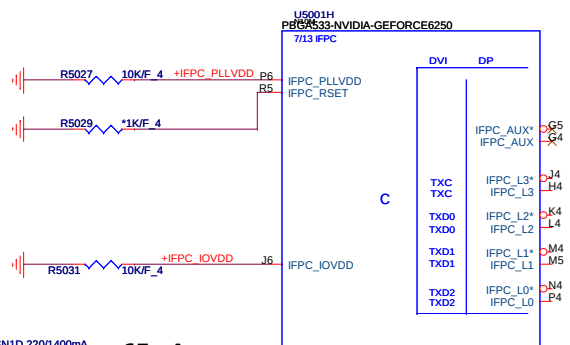
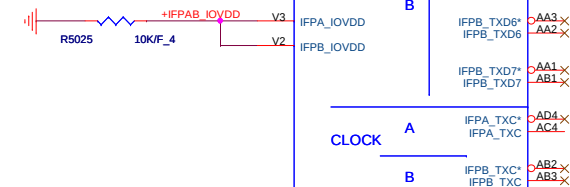
PROJECT : R13J
Quanta Computer Inc.

Size Custom Document Number N11M-GE2(PCIE/F) Rev 1A
Date: Wednesday, September 28, 2011 Sheet 14 of 38

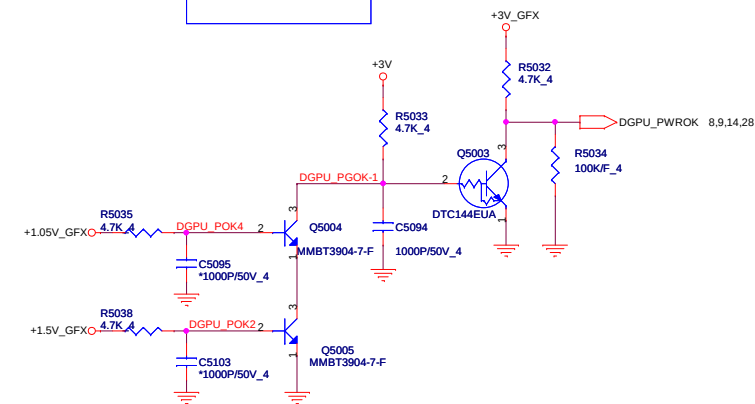
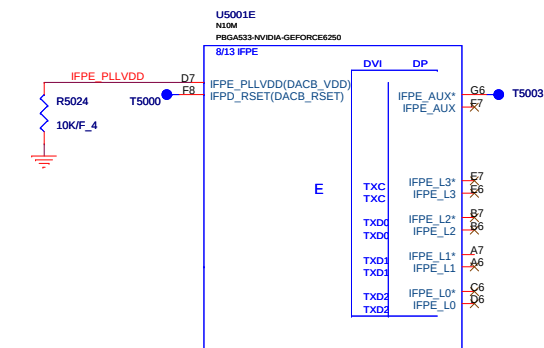
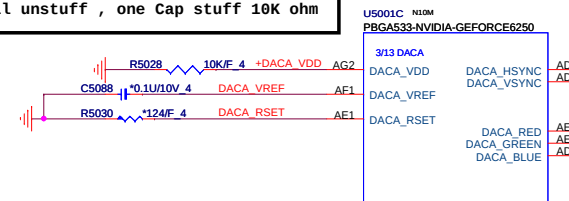


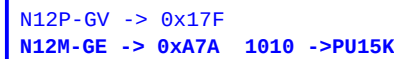


Optimus:
All unstuff , one Cap stuff 10K ohm

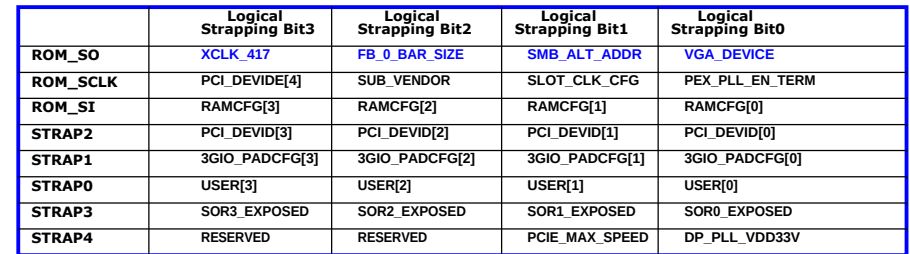


Optimus:
All unstuff , one Cap stuff 10K ohm

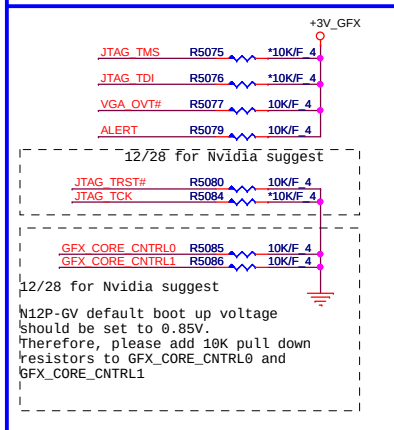




4.99K/F 4: CS24992FB26 [RES CHIP 4.99K 1/16W +1%(0402)]
10K/F 4: CS31002FB26 [RES CHIP 10K 1/16W +1% (0402)]
15K/F 4: CS31502FB24 [RES CHIP 15K 1/16W +1% (0402)]
30.1K/F 4: CS33012FB18 [RES CHIP 30.1K 1/16W +1%(0402)]
35.7K/F 4: CS33572FB13 [RES CHIP 35.7K 1/16W +1% (0402)]
45.3K/F 4: CS34532FB18 [RES CHIP 45.3K 1/16W +1% (0402)]



RAMCFG [3:0]	DESCRIPTION	Vendor	Vendor P/N	QCI P/N	ROM_SI
0000		Reserved			
0010	DDR3 64Mx16x8, 128bit, 1GB,800MHz	Hynix			PD 15K
0011	DDR3 64Mx16x8, 128bit, 1GB,800MHz	Samsung			PD 20K
0110	DDR3 128Mx16x4, 128bit, 1GB,900MHz	Hynix	H5T02G638FR-11C	AKD5MGWTW07	PD 35K
0111	DDR3 128Mx16x4, 128bit, 1GB,900MHz	Samsung	K4W2G1646C-HC11	AKD5MGWT508	PD 45K
XXXX					
XXXX					

[illegible]

GPI0	I/O	ACTIVE	USAGE
0	N/A	N/A	I-C-T
1	IN	N/A	Hot plug detect for IFP link C
2	OUT	HIGH	PANEL BACKLIGHT PWM
3	OUT	HIGH	PANEL POWER ENABLE
4	OUT	HIGH	PANEL BACKLIGHT ENABLE
5	OUT	N/A	NVVDD VID0
6	OUT	N/A	NVVDD VID1
7	OUT	N/A	NVVDD VID2
8	I/O	LOW	OVERT
9	I/O	LOW	ALERT
10	OUT	N/A	Memory VREF SELECT
11	I/O	N/A	SLI SYNC0
12	IN	N/A	PWR_LEVEL
13	OUT	N/A	THERM_LOAD_STEP_DOWN
14	OUT	N/A	THERM_LOAD_STEP_UP

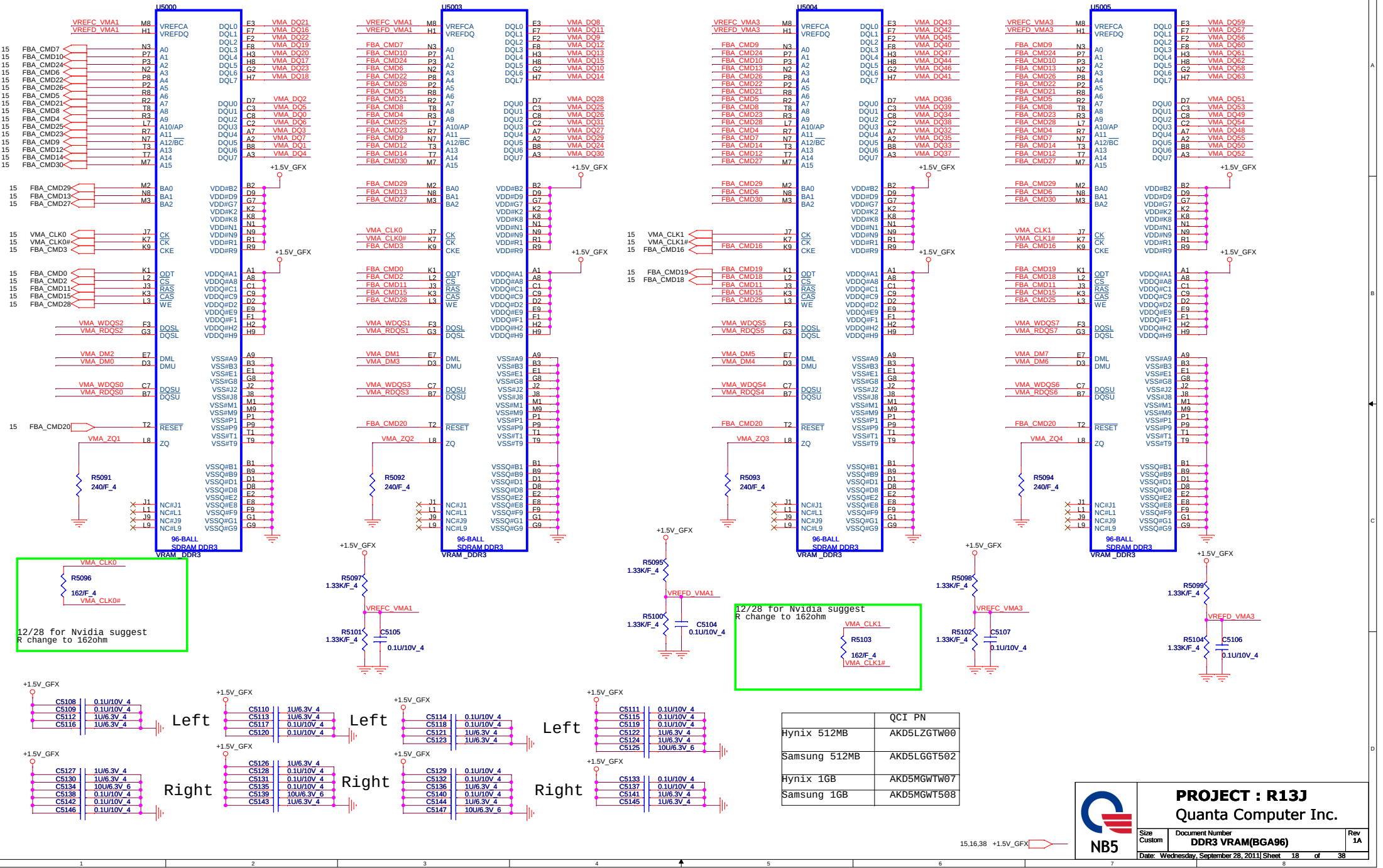


PROJECT : R13J
Quanta Computer Inc.

Size Custom	Document Number N12P-GV(GPIO/STRAPS)	Rev 1A
Date: Wednesday, September 28, 2011 Sheet 17 of 38		

CHANNEL A: 256MB/512MB DDR3

18

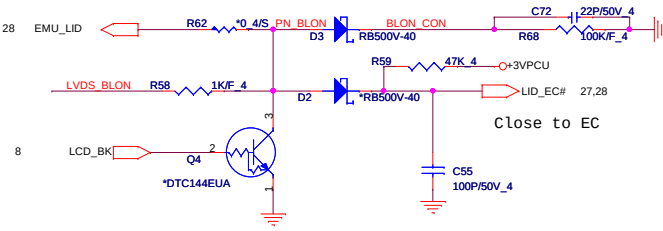


	QCI PN
Hynix 512MB	AKD5LZGTW00
Samsung 512MB	AKD5LGGT502
Hynix 1GB	AKD5MGWTW07
Samsung 1GB	AKD5MGWT508

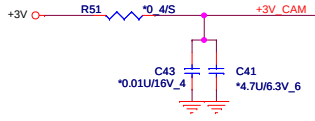
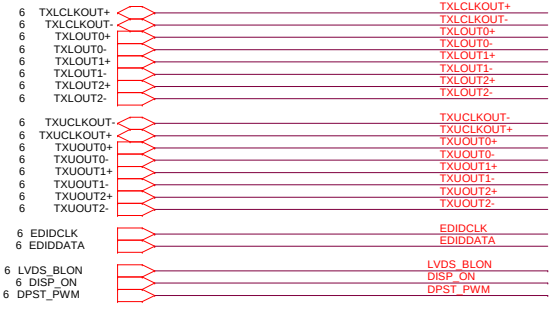
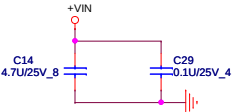
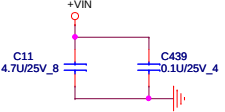
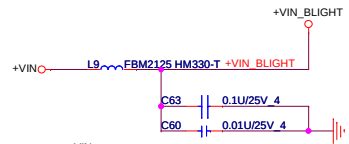
PROJECT : R13J
Quanta Computer Inc.

Size	Document Number	Rev
Custom	DDR3 VRAM(BGA96)	1A
Date: Wednesday, September 28, 2011 Sheet 18 of 38		

LID Switch



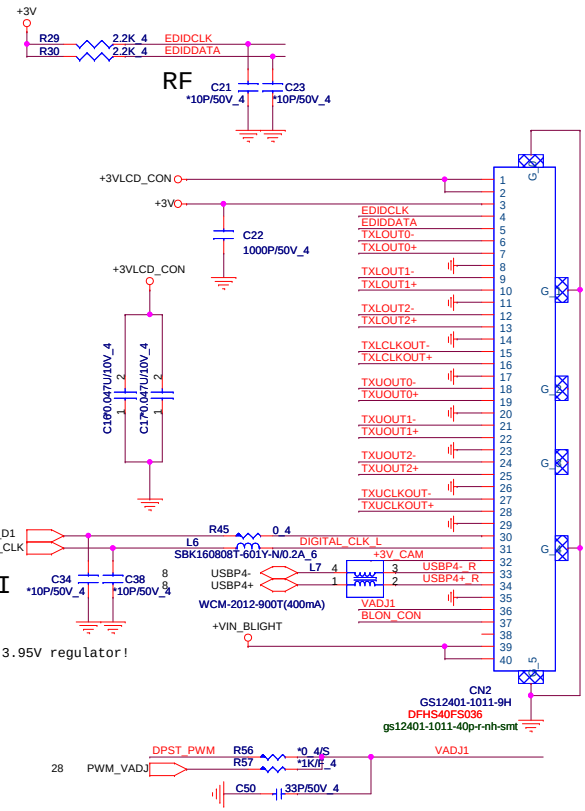
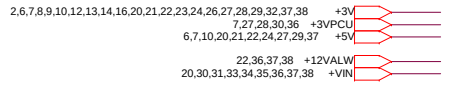
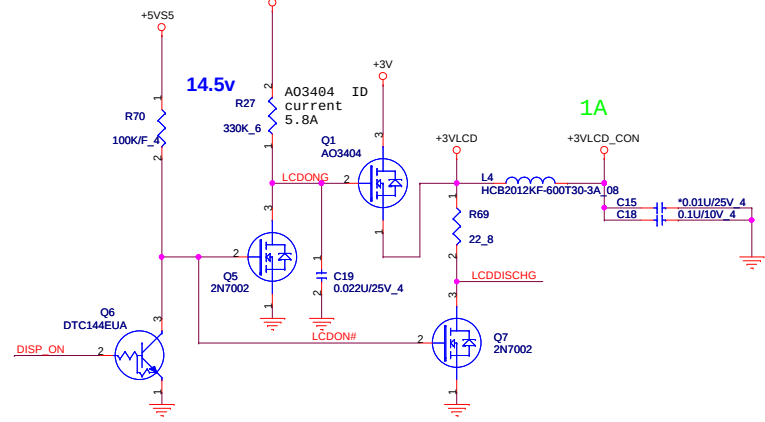
100mA

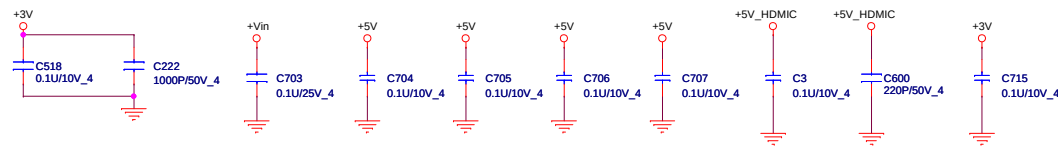


Please note that 2011 camera is +3V a We do not need to use 5V -> 3.95V regulator!

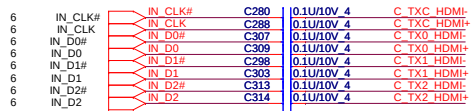
follow L7 location
DEL reserved Restor

change to +5VS5

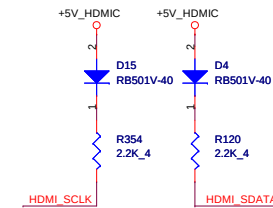
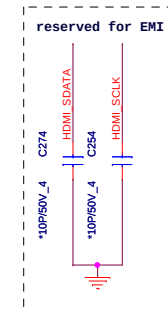
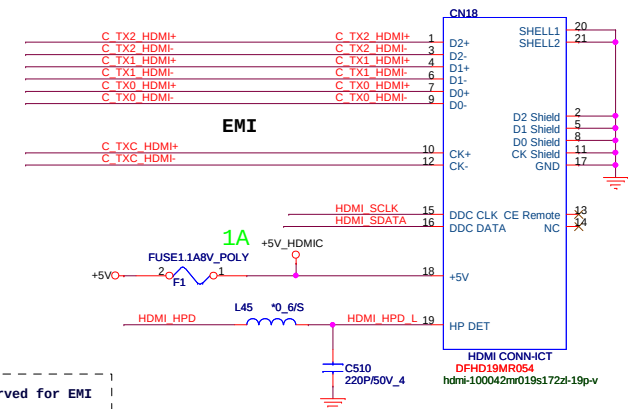
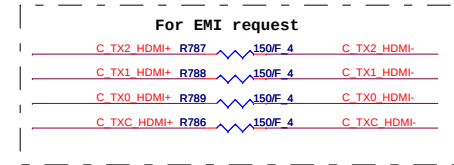
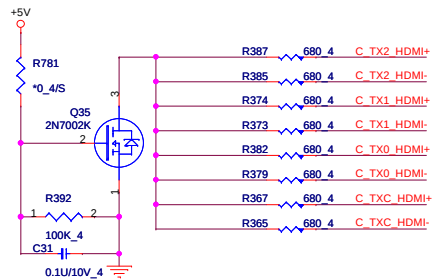
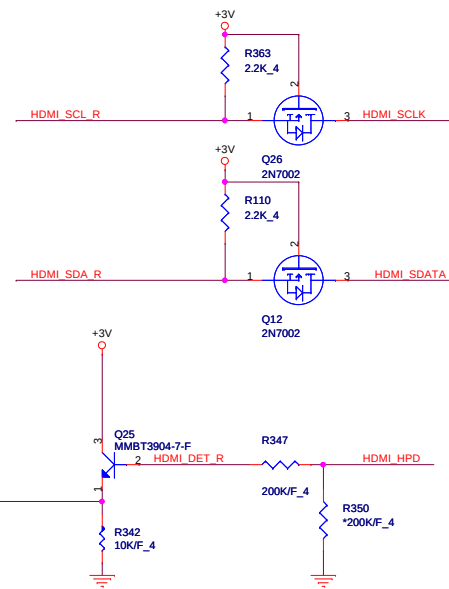


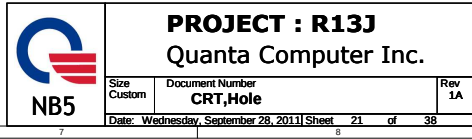


close to HDMI conn

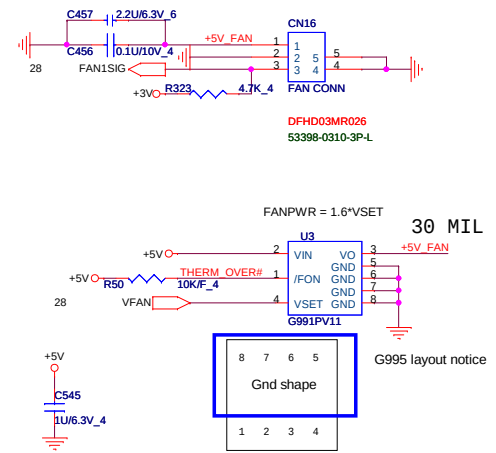


Close to HDMI Connector

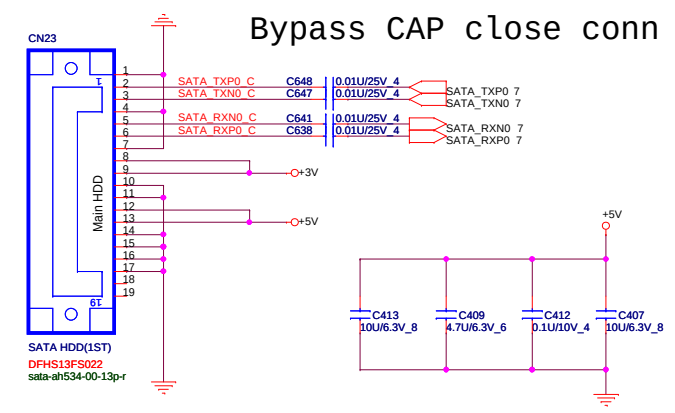




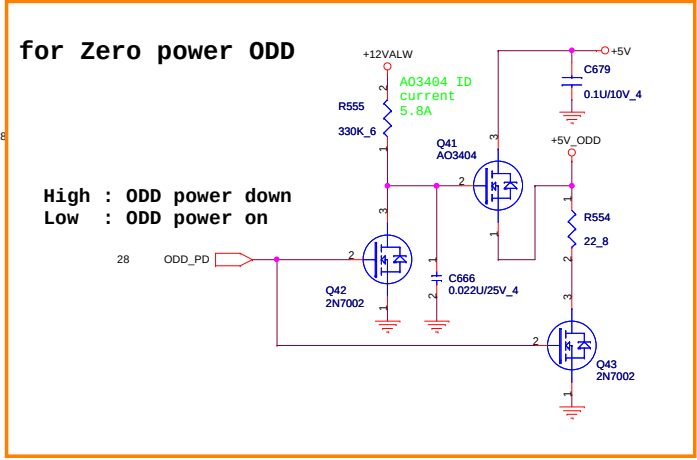
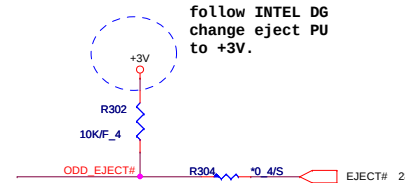
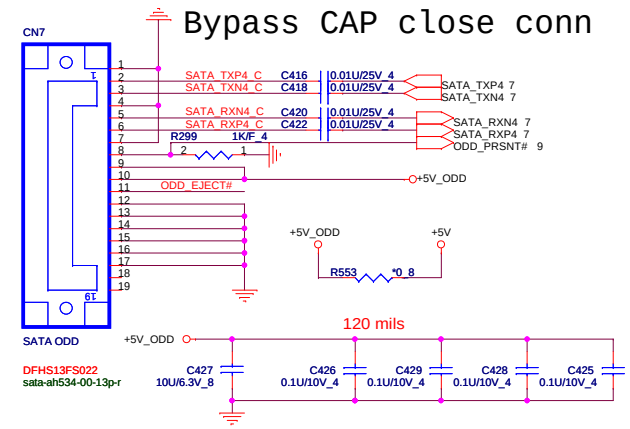
CPU FAN

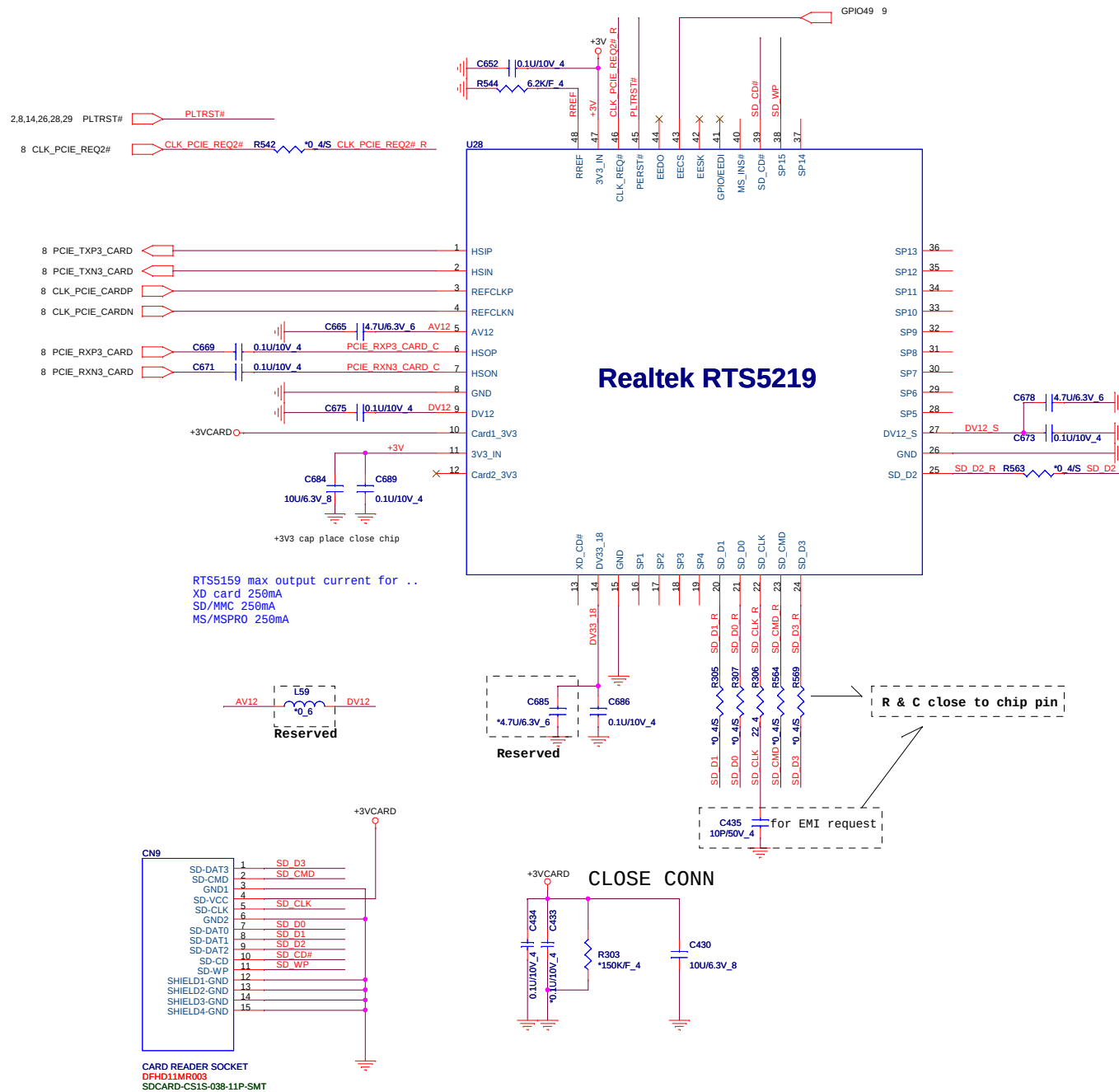


SATA HDD CONNECTOR

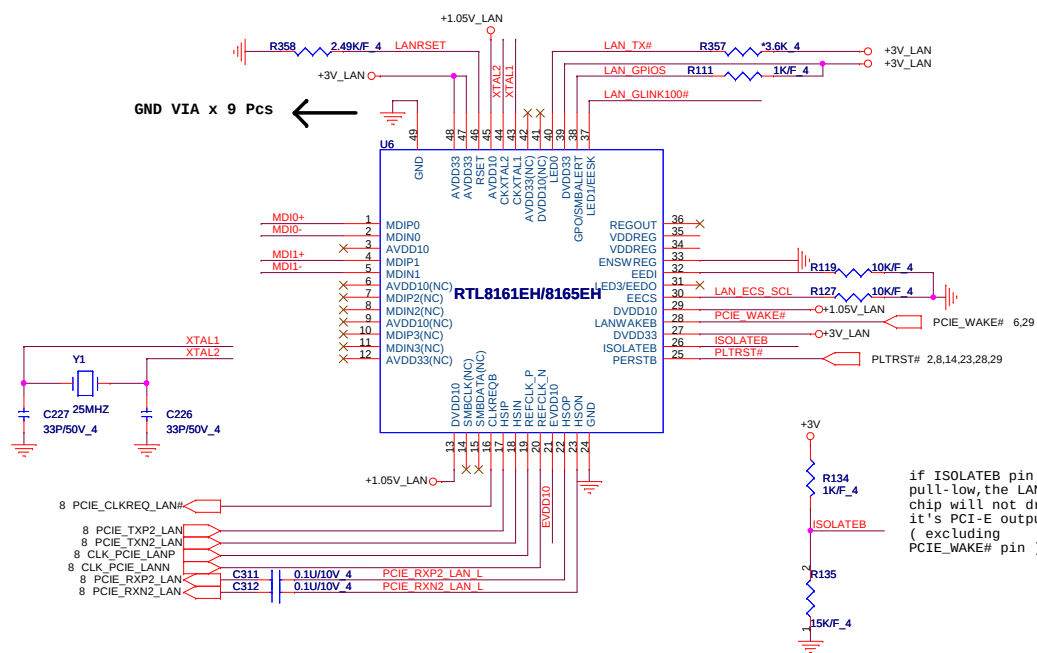


SATA ODD CONNECTOR

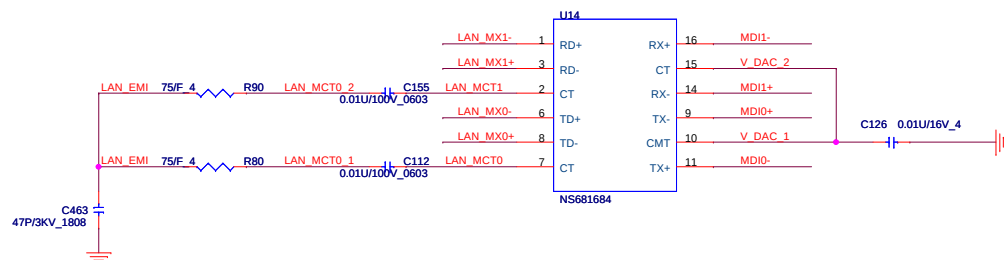




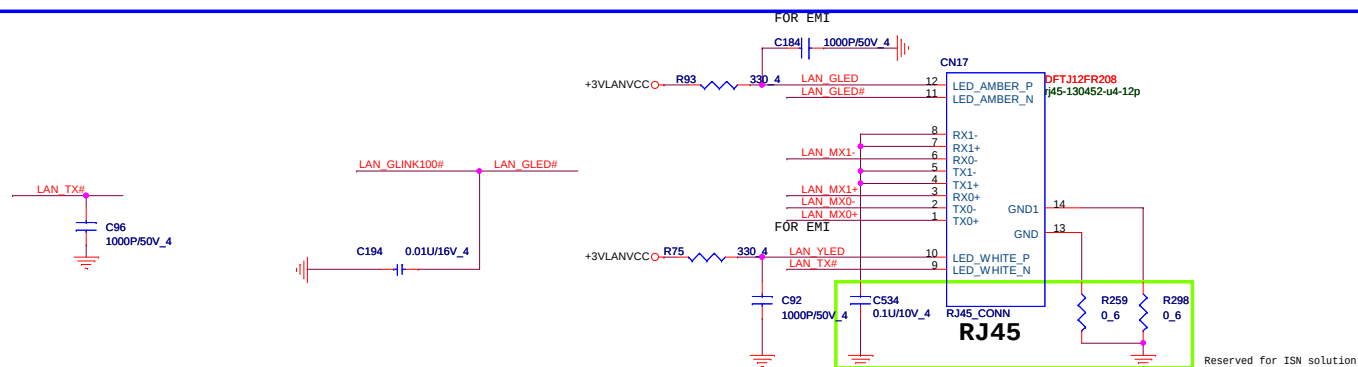




Transformer for 10/100

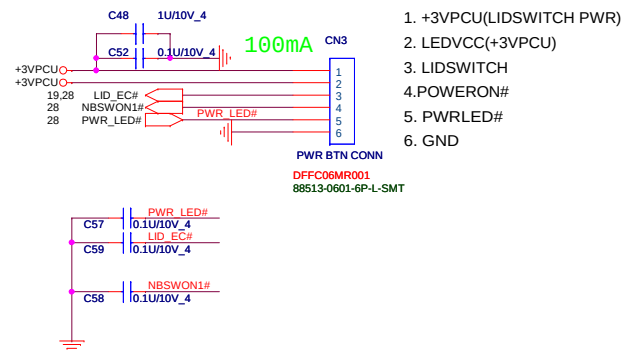


Lan Con.

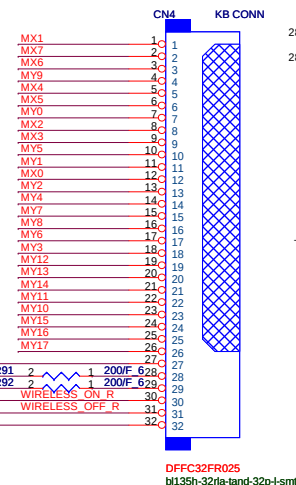
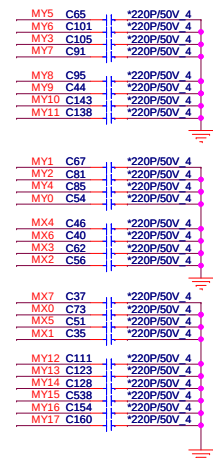


2, 6, 7, 8, 9, 10, 12, 13, 14, 16, 19, 20, 21, 22, 23, 24, 27, 28, 29, 32, 37, 38

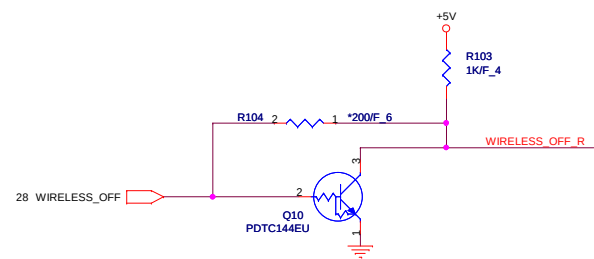
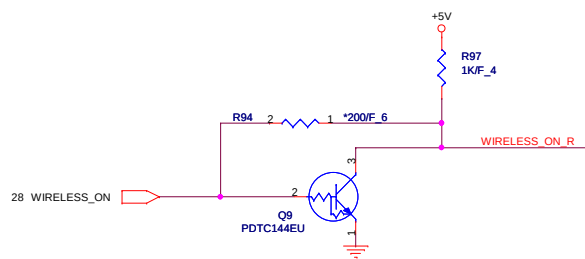
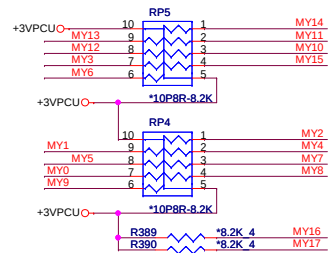
37 +3VLANVCC
+3V



KEYBOARD Con.

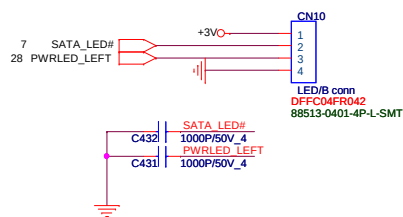


KEYBOARD PULL-UP



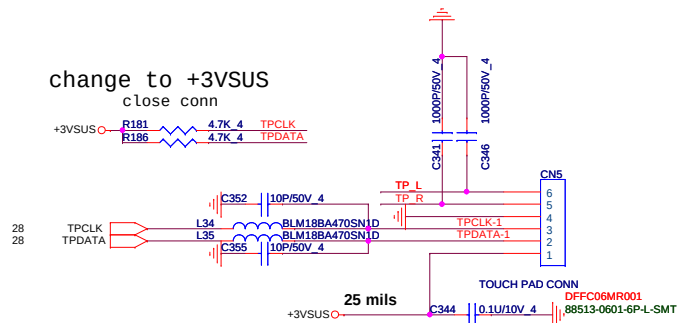
LED Con.

1. +3V
2. SATA_LED#
3. PWR_LED#
4. GND

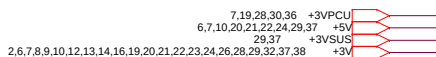
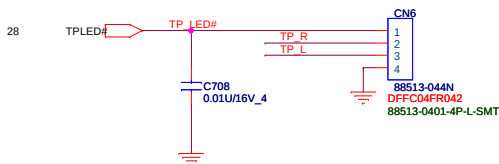


TOUCH PAD Con.

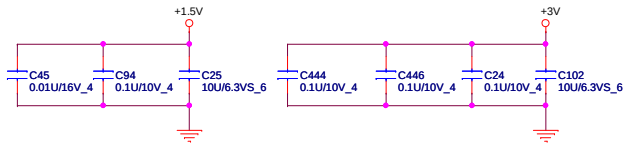
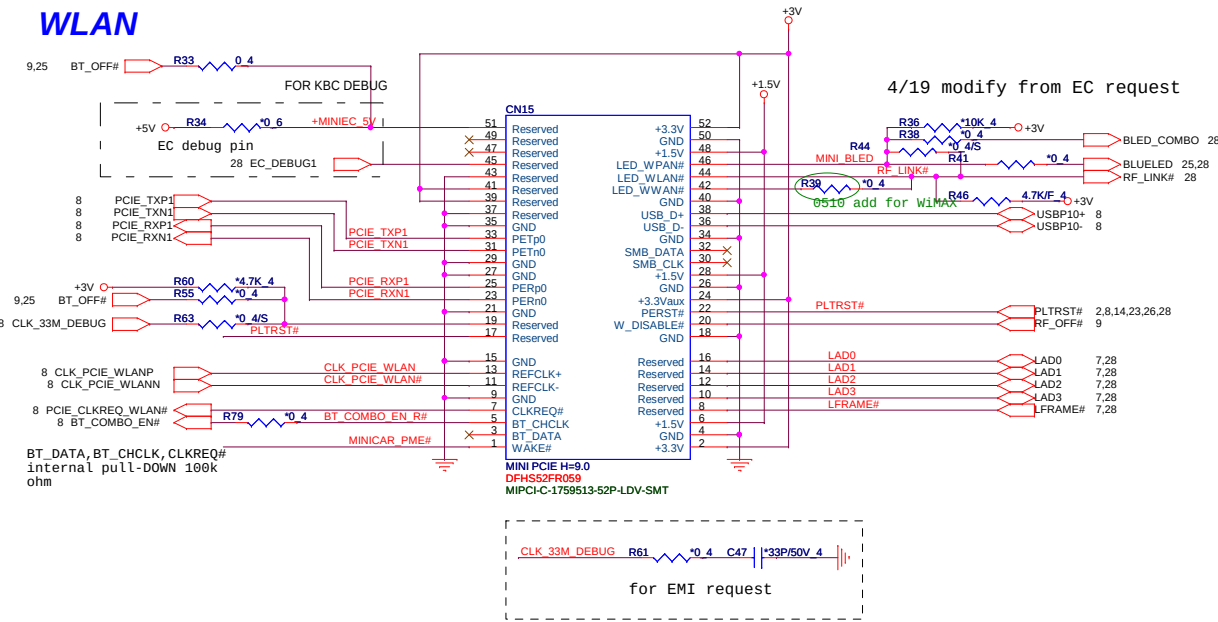
```
change to +3VSUS
      close conn
```



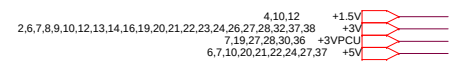
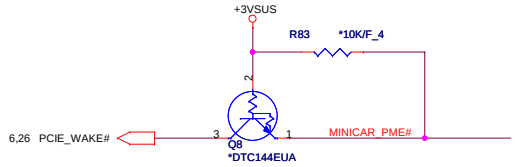
To TOUCH PAD SW board

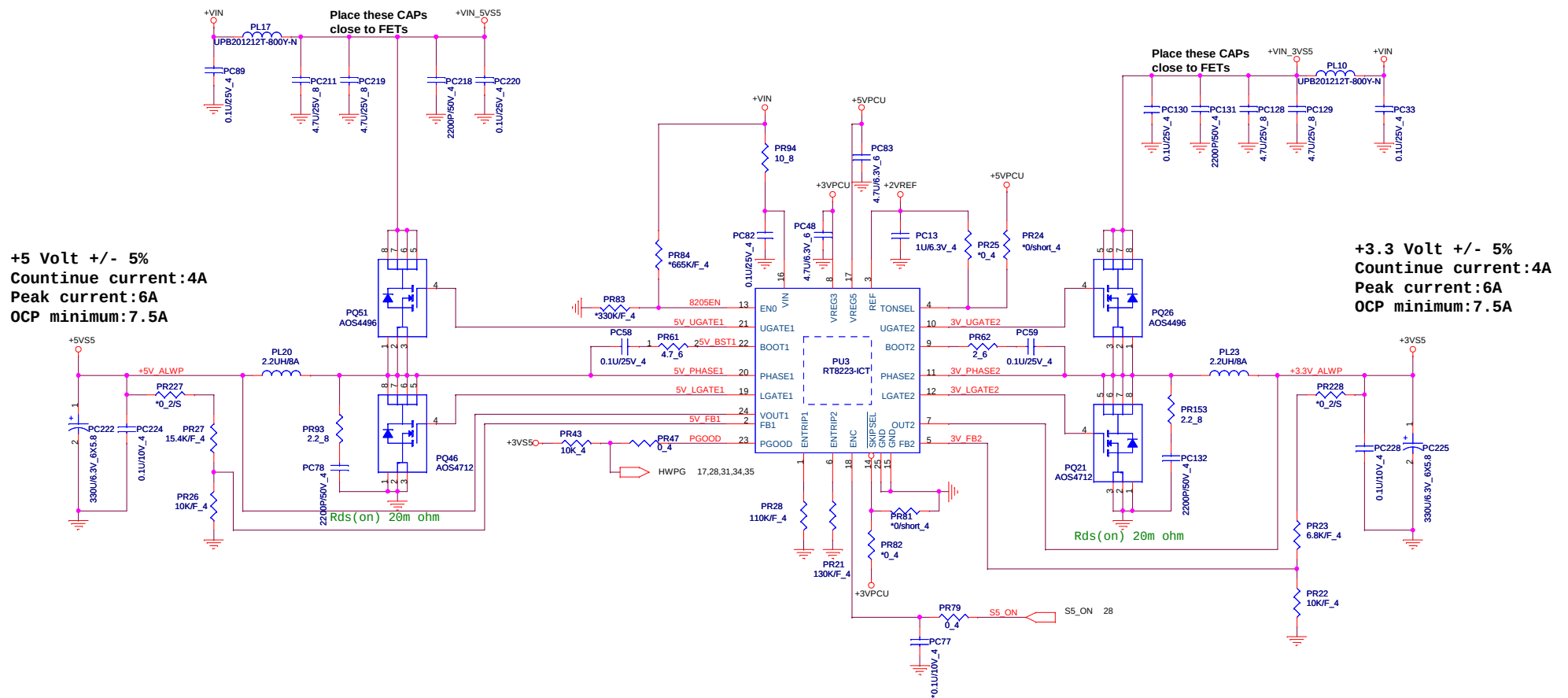


Mini PCI-E Card 1
WLAN

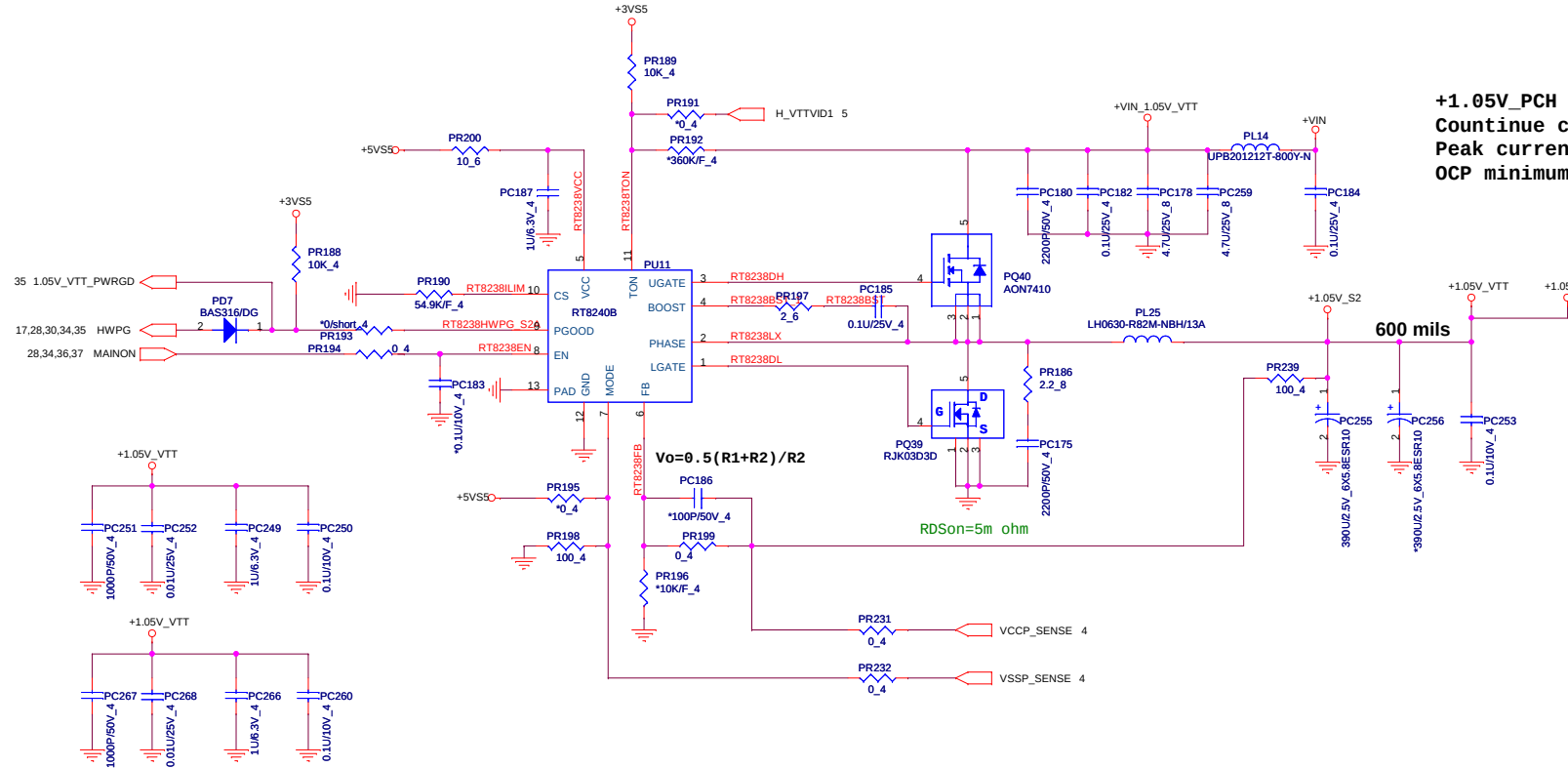


INTEL WLAN
CARD PIN 20
W_DISABLE#
have
internal
pull-up 110k
ohm

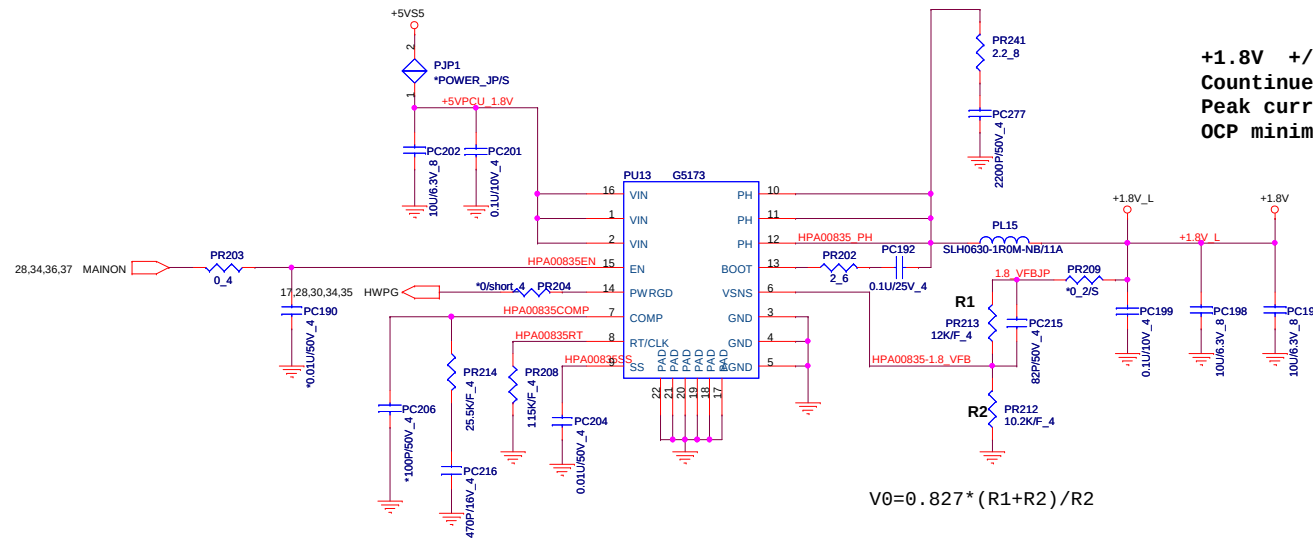


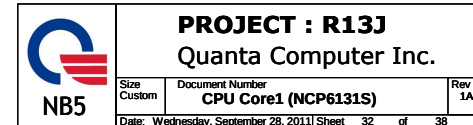


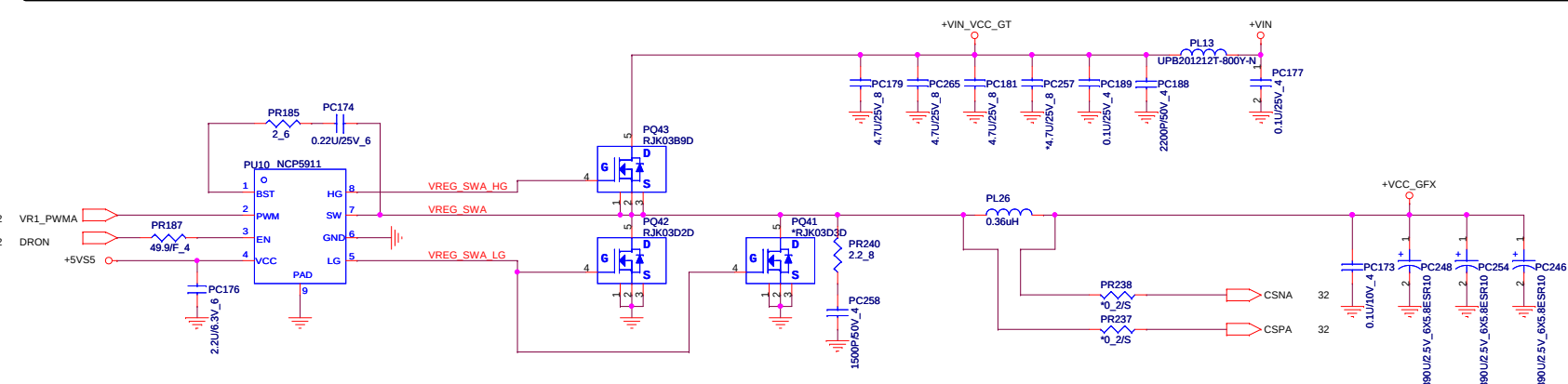
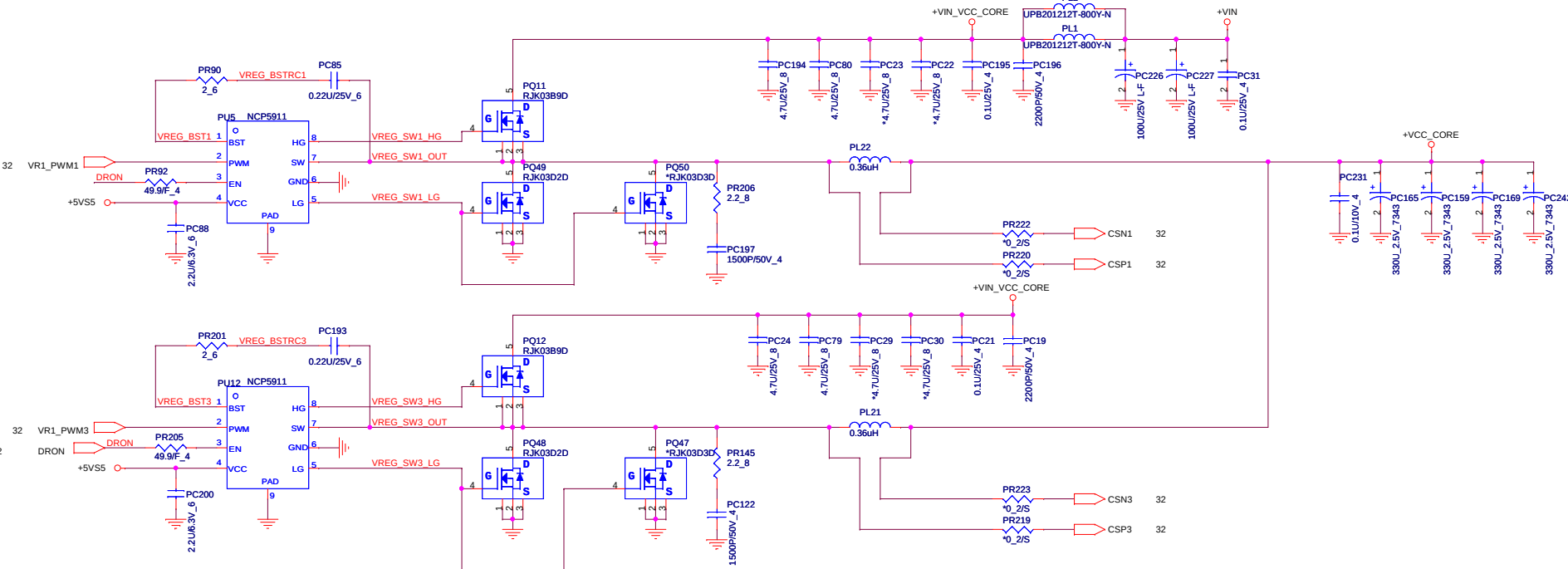
+1.05V_PCH Volt +/- 5%
Countinue current:10A
Peak current: 12A
OCp minimum: 14.5A

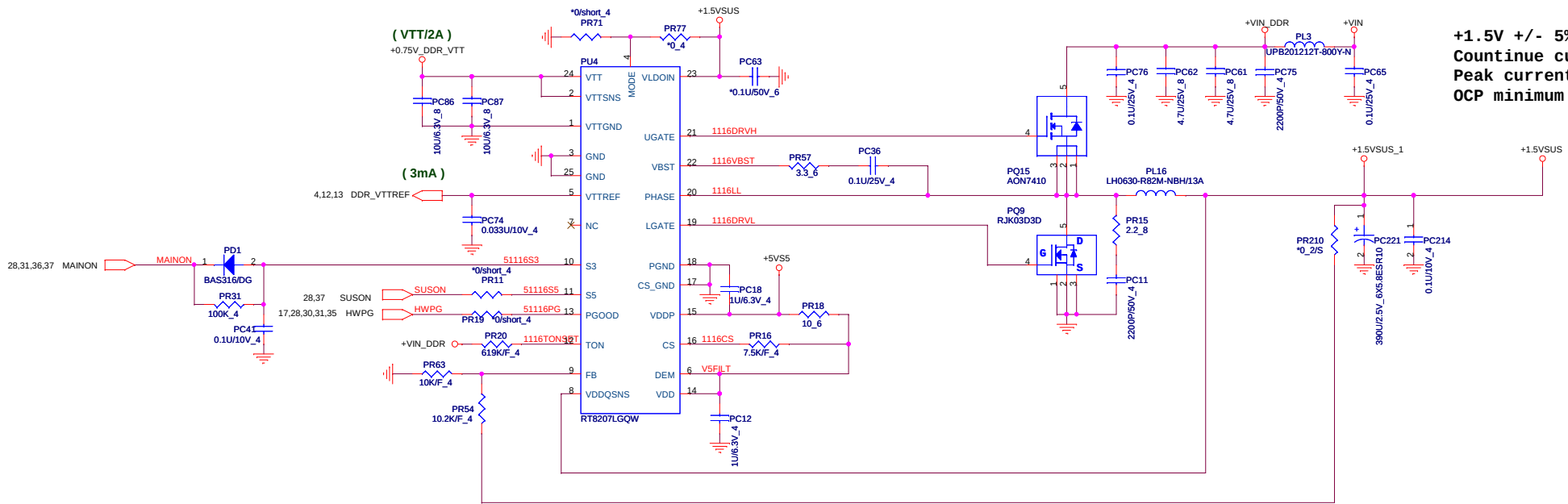


+1.8V +/- 5%
Countinue current:1.5A
Peak current:2.5A
OCp minimum 4A









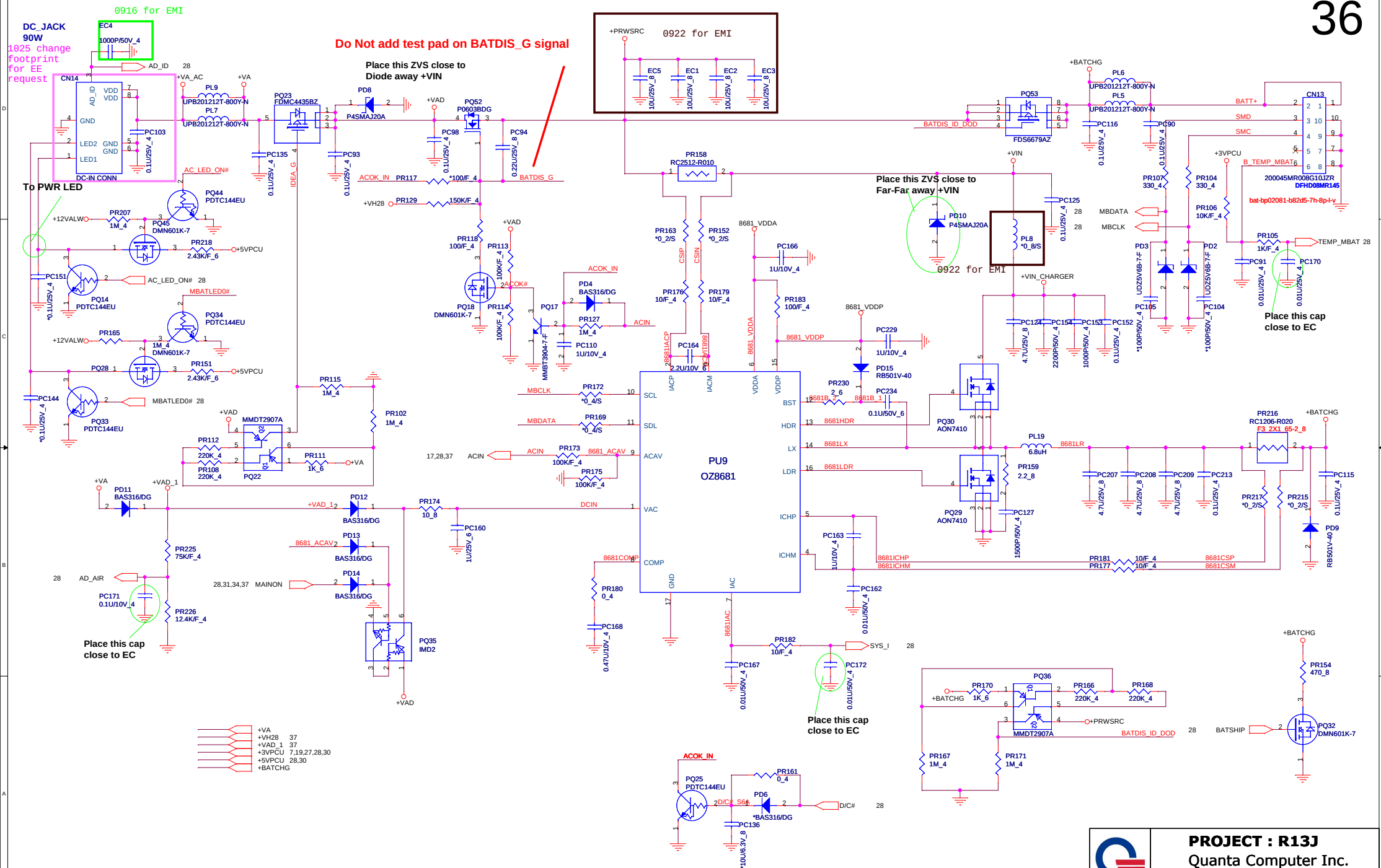
+1.5V +/- 5%
Countinue current:6A
Peak current:12A
OCP minimum 15A

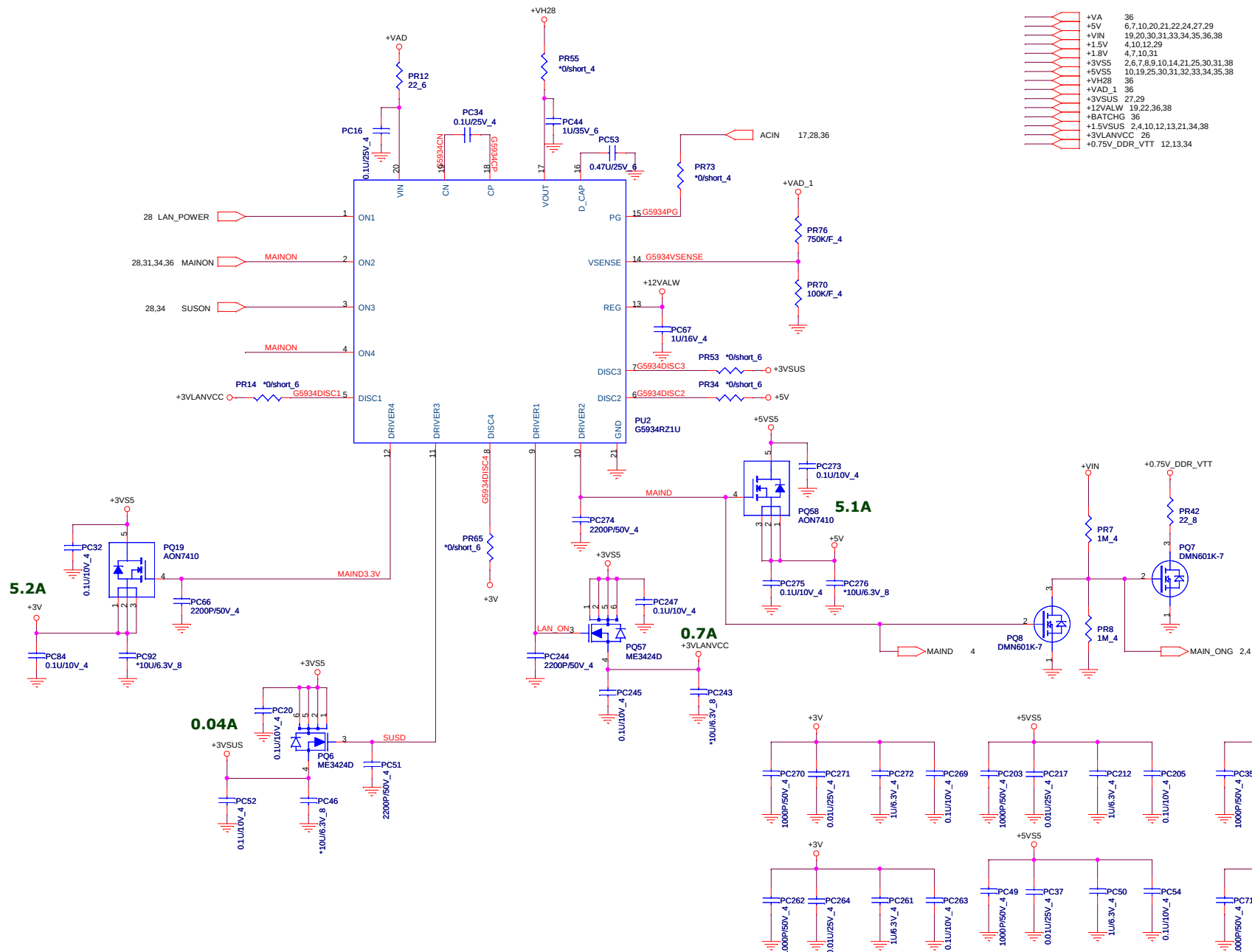
2011_0217 modify for EE request cancel 1.0V_VGA





200 mils





38

